

Who Bears the Cost? Automation and the Transmission of Monetary Policy

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Abstract

Automation makes disinflation cheaper in output terms, and it does so by shifting the cost towards the households least able to absorb it. We develop a Two-Agent New Keynesian model in which robotic capital is predetermined within the period of a monetary shock and enters a nested constant-elasticity technology as a close substitute for unskilled labour, with households partitioned separately by skill and by asset market participation. Predeterminacy leaves two margins free. The rental rate on a stock that cannot move must clear its market, and its weight in marginal cost rises with the robot cost share, which steepens the Phillips curve. Unskilled hours must absorb the whole movement in the composite input, in a proportion given in closed form by the same cost share, which concentrates the burden on constrained households. One parameter therefore governs both how much cheaper disinflation becomes and who pays for it. The aggregate result is a production-side one, surviving in an economy with no constrained households at all.

Keywords: automation capital, monetary policy transmission, Phillips curve, limited asset market participation, skill heterogeneity

JEL Codes: E25, E31, E52, O33, J24

1 Introduction

The world operational stock of industrial robots rose from roughly 1.47 million units in 2014 to 4.66 million in 2024, growing by a further nine per cent in the most recent year alone, while annual

*Replication code is available at https://github.com/alvaromezquita/robot_tank_model.

installations doubled over the same period, from 221,000 to 542,000 units, and are projected to approach 708,000 by 2028 ([International Federation of Robotics, 2025](#)). Figure 1 plots the two series. Accumulation on this scale invites comparison with earlier industrialisations, and the comparison is instructive mainly for the way it fails. The machinery of the steam age and of the railway age was, in the main, machinery that made workers more productive at what they were already doing: it raised the value of an hour of labour, and its accumulation raised the demand for labour along with it. This is the presumption that macroeconomics has inherited and that its aggregate production functions encode, in which capital and labour are complements and capital deepening is good for wages. Robotic capital is defined by the opposite property. It does not make a worker faster at a task; it performs the task instead of her. The difference is not one of degree along a spectrum of labour-saving machinery, because the two kinds of capital enter the demand for labour with opposite signs: accumulating the first kind raises it, accumulating the second withdraws work from the people who did it. A decade of accumulation at the rate reported above is therefore not simply more capital. It is a shift in the composition of the capital stock towards the kind whose effect on the demand for labour runs the wrong way.

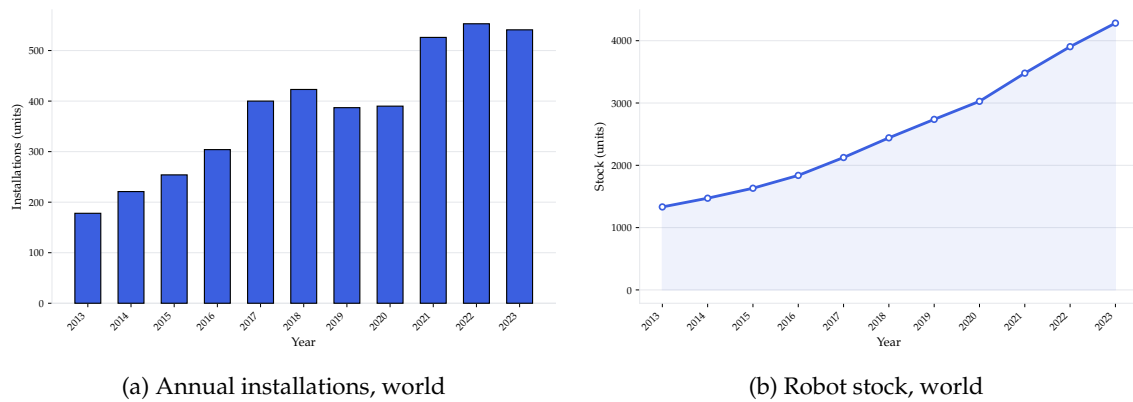


Figure 1: Evolution of robots in the last years in 1,000 units ([International Federation of Robotics, 2025](#))

This shift has two consequences, and monetary policy meets both. The first is aggregate. A central bank acts on inflation by acting on the cost of production, and the cost of production responds to the policy rate through the inputs firms adjust in the short run. As the share of production accounted for by an input that substitutes for labour grows, the responsiveness of costs to a policy move is governed by a different mix of margins than before, and the terms on which the central bank can trade output against inflation need not be the terms it faced in a less automated economy. The second is distributional. The tasks that robots absorb were performed by particular workers, and the robots absorbing them are owned by particular households, so a shift in the composition of production is also a shift in who is exposed to a shock and who is insulated from it. Whether these two consequences pull in the same direction, and what follows for policy if they do, is what this paper asks.

Not all capital automates, and the distinction is one of kind rather than of degree. Following [Acemoglu and Restrepo \(2022\)](#), we take automation to mean a technology that expands the set of tasks

performed by capital, displacing workers from tasks in which they previously held a comparative advantage. A factor-augmenting technology raises a worker’s productivity at the tasks she already performs, and whether it raises or lowers her relative wage depends on the elasticity of substitution between her labour and other factors; an automating technology reallocates the task itself, and the resulting effect on her wage is negative whatever that elasticity turns out to be. Automation capital is accordingly defined here by function and not by form: it is that part of the capital stock whose distinguishing property is an unusual degree of substitutability for labour in routine tasks, work reducible to a well-specified set of procedures, alongside complementarity with the labour that performs the rest. Robotic capital is the subset of that class which performs such tasks in the physical world; the automation of routine cognitive tasks lies outside our scope. Defining the object by function leaves open what is to be counted as an instance of it, and here we adopt the convention that underlies the measurements reported above. The international standard defines a robot as an actuated mechanism, programmable in two or more axes, with a degree of autonomy, that moves within its environment to perform intended tasks (ISO, 2021). Two elements of that definition matter economically rather than technically. Programmability is what makes a single unit reassignable across tasks, so that the class of work exposed to it is not fixed by the design of any one machine; and autonomy is what allows the unit to execute a task rather than to assist a worker executing it, which is the functional criterion above restated in engineering terms. The correspondence is not incidental: the stock series in Figure 1 counts exactly the units meeting this standard, so the empirical object and the theoretical one refer to the same thing.

We restrict attention to this subset for two reasons. The first is measurement: it is the only part of the class with a separately observed stock, and the series of Figure 1 has no counterpart for the remainder. The second is that the physical character of the task is what makes the substitution asymmetric across skill types, which is the one property the mechanism developed below actually requires. That property can be stated exactly: the elasticity of substitution between robotic capital and unskilled labour must exceed the elasticity between robotic capital and skilled labour. This is a statement about relative substitutability rather than about the level of either elasticity, the direction of employment shares, or the shape of the occupational distribution, and it is one that has been estimated directly (Battisti et al., 2021).

The asymmetry is visible in the data on incidence. Robot adoption has lowered the employment share of low-skilled workers relative to more skilled ones and depressed employment and wages in local labour markets exposed to it, with the effect concentrated in routine and manual occupations (Graetz and Michaels, 2018; Acemoglu and Restrepo, 2020). One feature of that evidence follows into the model. Substantial movements in the wage structure are found to coexist with only modest aggregate productivity gains (Acemoglu and Restrepo, 2022), so the distributional consequence of automation is not the by-product of a first-order efficiency gain that might be presumed to compensate for it.

The two consequences have been studied apart. The frameworks that give automation capital an explicit role in New Keynesian transmission (Lin and Weise, 2019; Fueki and Maehashi, 2019; Basso and Rachedi, 2025) are built on a representative household and are therefore structurally unable to speak to incidence, while the literature on the distributional consequences of monetary policy (Coibion et al., 2017; Kaplan et al., 2018; Dolado et al., 2021) attaches its households to no

automation technology. Where the two are brought together, it is outside the monetary setting: [Jaimovich et al. \(2021\)](#) and [Guerreiro et al. \(2022\)](#) both pair automation capital with households differing in their exposure to it, but neither has a nominal side or a policy rate, and both ask what a planner should do about automation rather than what automation does to the transmission of a policy the central bank is already conducting. Section 2 develops the comparison in full.

Our motivation is to bring the two together. Specifically, we aim to address two inquiries:

1. How does the degree of automation alter the transmission of monetary policy to output and inflation?
2. How does automation alter the incidence of that transmission across household types?

To address these questions, we construct a Two-Agent New Keynesian model following [Bilbiie \(2008\)](#), augmented with robotic capital as a productive input in a nested constant-elasticity-of-substitution technology in the manner of [Fueki and Maehashi \(2019\)](#). Households are partitioned along two dimensions rather than one: by skill type and by participation in asset markets, so that skilled and unskilled workers may each be Ricardian or hand-to-mouth. Robotic capital enters the same nest as unskilled labour, as a close substitute for it, while skilled labour enters the upper nest, and the stock of robots is predetermined with respect to the period in which the monetary shock arrives. Unlike the steady-state comparative statics that a static treatment permits, the predetermined stock adds a state variable that precludes a closed-form reduction, so the equilibrium dynamics are obtained numerically and the impulse responses to a monetary policy shock, computed across economies differing in their degree of automation, are our object of study.

The paper reports one mechanism and three readings of it. Automation makes disinflation cheaper in output terms: the output cost of a given disinflation declines monotonically as the robot share of production grows. The mechanism is predeterminacy. With the robot stock unable to move within the period of the shock, a larger robot share means a larger fraction of the short-run adjustment in marginal cost must fall on the margins that remain free, so inflation carries more of the response and output less. One of those margins is unskilled hours, supplied disproportionately by households who cannot smooth, so the same rigidity that makes disinflation cheaper in the aggregate concentrates its cost: the consumption loss of unskilled hand-to-mouth households rises steadily relative to that of savers as automation deepens, while the loss borne by savers is flat across regimes. The third reading is that the aggregate result owes nothing to limited asset market participation, against the natural expectation for a two-agent model. In a counterfactual economy with no constrained households at all, the output cost of disinflation still declines at essentially the same rate and the unskilled wage still deepens faster than the skilled. The steepening of the Phillips curve is a production-side result; the asset market constraint governs the level of the distributional cost rather than its gradient.

Two parameters carry the distributional claim, and neither is disciplined by the steady state. The sign of the constrained households' hours response turns on whether relative risk aversion lies

below unity, and the scale of the skill gap turns on the elasticity of substitution between robots and unskilled labour, which enters the dynamics without leaving any trace in the steady-state shares the calibration targets. We report results across a grid in both rather than defending point values. The direction of every claim above holds at every value of either parameter that we consider; the magnitude of the distributional claim does not, and Appendices C.2 and C.3 report how far it moves.

The reading for policy is positive rather than prescriptive. Nothing here says that a central bank should respond differently to automation. What it says is that the trade-off facing a central bank in an automating economy is not the one it faced before, that the apparent improvement in that trade-off is not free, and that the cost is borne by precisely the households its instrument reaches last and least. Whether that warrants a response, and through which instrument, is a question about the assignment of instruments to objectives rather than about monetary policy alone; Section 5 sets out what the model does and does not license on this point.

The remainder of the paper is organised as follows. Section 2 reviews the literature on automation, household heterogeneity and the transmission of monetary policy, and positions our contribution within it. Section 3 sets out the model. Section 4 presents the calibration, the experimental design, and the aggregate and distributional responses to a monetary policy shock across economies differing in their degree of automation. Section 5 draws out the implications for policy. Section 6 concludes. Derivations, the computational specification and the sensitivity analysis are collected in the appendices.

2 Literature review and discussion

Robotic capital is increasingly being adopted as a substitute for routine and unskilled labour tasks. The hypothesis that technology displaces workers task by task rather than skill by skill originates with Autor et al. (2003), who argue that computerisation substitutes for labour in codifiable routine work and complements it elsewhere, and Goos et al. (2014) estimate a framework in that spirit across sixteen Western European countries, attributing the hollowing-out of the middle of the occupational distribution to routine-biased technological change and offshoring. Evidence specific to industrial robots points the same way: Graetz and Michaels (2018), using an industry panel across seventeen countries, find that robots did not reduce total employment but did lower the employment share of low-skilled workers relative to middle- and high-skilled ones, while Acemoglu and Restrepo (2020) estimate robust negative effects of robot exposure on employment and wages across US local labour markets. The picture is not uncontested. Oesch and Piccitto (2019) document occupational upgrading rather than polarisation in Germany, Spain, Sweden and the United Kingdom over two decades, with employment growth concentrated in the better-paid occupations rather than at both ends of the distribution, so the shape of the occupational structure under automation remains an open empirical question.

That dispute bears less on what follows than it might appear, because the mechanism developed below does not rest on the shape of the occupational distribution. What it requires is that robotic capital be more readily substitutable for unskilled than for skilled labour, which is a statement about relative elasticities of substitution rather than about the trend in employment shares, and one that can be assessed directly. [Battisti et al. \(2021\)](#), using a constructed measure of the robotic capital stock across countries and industries, estimate a lower elasticity of substitution between robots and skilled labour than between robots and unskilled labour, and find this confirmed on both of their samples and under alternative constructions of the stock. Their finer partition qualifies the result: once middle-skilled workers are separated out, they emerge as the group most exposed to robotisation, so the substitutable margin sits nearer the middle of the distribution than its bottom. The two-type structure adopted here collapses that distinction, and the unskilled household should accordingly be read as supplying routine and codifiable labour rather than as occupying the lowest part of the wage distribution. This asymmetric incidence is what motivates both our restriction of the term “robotic capital” to this class of automation and the production structure developed below. We treat the degree of substitutability as a quantity the results are conditional on, and report their sensitivity to it in [Appendix C.2](#). Against this backdrop, several studies have examined how automation alters inflation dynamics and the transmission of monetary policy. However, none of these studies has considered how these effects interact with income inequality driven by skill heterogeneity in the labour market.

For household heterogeneity, we build on the two-agent model of [Bilbiie \(2008\)](#), which distinguishes between two representative households: one exhibiting Ricardian behaviour, which consumes and saves, and another exhibiting hand-to-mouth behaviour, which consumes all current income. The share of hand-to-mouth households is shown to be crucial for aggregate dynamics, and the mechanism runs through the cyclicity of the constrained households’ income: when their income responds more than proportionally to aggregate income, aggregate-demand shocks are amplified, and for a sufficiently strong response the slope of the IS relation can change sign. [Bilbiie \(2025\)](#) later generalises this two-agent structure into an analytical heterogeneous-agent framework in which this income elasticity becomes the model’s sufficient statistic, showing that countercyclical inequality amplifies aggregate demand. This framework naturally lends itself to our analysis because it incorporates heterogeneity across households, which we extend by introducing differentiation in labour skills (see also [Galí et al., 2007](#), for a related two-agent structure applied to fiscal policy).

This income-elasticity channel has more recently been grounded empirically. [Auclert \(2019\)](#) recasts it as a transmission channel of monetary policy in its own right: because the households whose income rises most after a monetary expansion also tend to hold the highest marginal propensities to consume, redistribution amplifies aggregate spending—an “earnings heterogeneity” channel that he expresses as an estimable sufficient statistic rather than a structural parameter. [Patterson \(2023\)](#) measures this object directly in US administrative data, documenting a large, positive covariance between households’ marginal propensities to consume and the cyclical sensitivity of their earnings—sufficient, she estimates, to raise the amplification of demand shocks by roughly forty percent relative to a benchmark of equal incidence. In this literature, however, the cyclicity of the constrained households’ income is taken as given: shaped by fiscal redistribution or by the incidence of labour-market shocks, or recovered as a reduced-form statistic, but never de-

terminated by the structure of production. It is precisely there that our mechanism operates—the substitutability between robotic capital and unskilled labour makes the income elasticity of the constrained households an endogenous object, responding to the degree of automation.

Closest to our structure is [Bilbiie et al. \(2022\)](#), who augment this two-agent framework with physical capital. They identify a complementarity between capital inequality—only savers hold the productive asset—and income inequality: because capital income is procyclical, an expansion redistributes it toward the constrained households, who consume it and set off a further round of demand, amplifying the response of aggregate consumption (the “multiplier of the multiplier”), and they show that this procyclical investment forces the central bank to respond more aggressively to inflation to preserve equilibrium determinacy. Their capital, however, is a generic productive asset, complementary to labour and held only by savers. Our robotic capital differs precisely on this margin: it substitutes for the unskilled labour supplied by the constrained households, so the factor-income channel underlying their amplification is reshaped when the asset displaces, rather than complements, the constrained agent’s income.

A parallel argument has been made outside the New Keynesian setting. [Moll et al. \(2022\)](#) show that once the long-run supply of capital slopes upwards rather than being perfectly elastic, automation raises the return to wealth permanently instead of being absorbed into quantities, so that the gains accrue to owners of the asset and average wages may stagnate even as output expands. The asset that displaces labour is held by the households whose incomes are already highest, and the distribution widens through returns rather than through the wage structure alone. Their object, however, is the secular drift of that distribution as automation is adopted, driven by the adoption itself. Ours is the cyclical incidence of a monetary shock at a given degree of automation, where the stock is predetermined and cannot respond over the horizon of transmission. The two are complementary: they establish ownership of automation capital as a channel of distribution in the long run, and we ask what that ownership implies for the transmission of policy in the short run.

In [Lin and Weise \(2019\)](#), robots are introduced as a distinct productive input in a NK model via a nested CES production function, along with traditional capital and labour. They show that robotic capital has effects on business cycles that are quantitatively distinct from those of traditional capital shocks, implying that monetary policy would need to place greater emphasis on output stabilisation. With a similar production structure, [Fueki and Maehashi \(2019\)](#) show that robotisation weakens the relationship between inflation and real activity, effectively flattening the Phillips curve. Lastly, [Basso and Rachedi \(2025\)](#) provide both empirical and theoretical evidence that robot adoption flattens the Phillips curve through the weakening of workers’ bargaining power. [Leduc and Liu \(2024\)](#) sharpen this channel by making the threat rather than the act of automation the operative force. In a search-and-matching model where a firm may fill a vacancy with a machine instead of a worker, the option to automate lowers the wage that can be bargained even in periods when no vacancy is converted, and because that option is worth more when labour markets are tight, it generates an endogenous real wage rigidity that amplifies unemployment fluctuations. Their extension to two skill types places robots alongside skilled workers in one intermediate good and unskilled workers in the other, so that automation raises the skill premium along the same margin we exploit. The mechanism nonetheless differs from ours in where the

short-run adjustment falls: their automation margin is endogenous and reaches the wage through the outside option in bargaining, whereas ours is a predetermined stock that reaches the wage through the conditional factor demands of a nested technology, with hours supplied rather than matched. With the exception just noted, these studies do not distinguish between skilled and unskilled labour, and none pairs that distinction with heterogeneity in asset market participation, so they cannot address how automation reshapes the distribution of income between households who own the automation capital and households who do not.

Most closely related to our setup is [Dolado et al. \(2021\)](#), who study the distributional consequences of monetary policy in a New Keynesian model with incomplete asset markets, skill-differentiated labour, and a nested-CES production function embedding capital-skill complementarity in the sense of [Krusell et al. \(2000\)](#). In their framework, capital is complementary to skilled labour and substitutable with unskilled labour, so that a monetary expansion—by raising both labour and capital demand—widens the skill wage gap, an effect reinforced by asymmetric search-and-matching frictions that favour skilled workers. Our model shares their skill-differentiated, nested-CES architecture but inverts the nesting: robotic capital enters the same nest as unskilled labour, as a close substitute for it, rather than being nested with skilled labour. The redistributive force therefore operates through direct factor substitution against the constrained households rather than through labour-market frictions. More importantly, their analysis characterises how a given monetary shock is distributed across worker types; it does not examine how the presence of automation reshapes the transmission of monetary policy itself, nor how it alters the incidence of that transmission across household types. It is precisely these questions that our framework is designed to address.

A distinct strand studies the distributional incidence of monetary policy—who gains and who loses when the central bank moves rates—rather than its aggregate effects. Empirically, [Coibion et al. \(2017\)](#) document that contractionary monetary shocks in the United States systematically widen inequality in labour earnings, income, and consumption, and account for a non-trivial share of its cyclical variation. Theoretically, [Kaplan et al. \(2018\)](#) show why this incidence requires household heterogeneity to appear at all: decomposing the consumption response to a rate change into a direct intertemporal-substitution effect and indirect general-equilibrium effects operating through labour income, they find the former dominates in a representative-agent economy but the latter dominates once realistic marginal propensities to consume are present—so that transmission to constrained households runs almost entirely through their labour income. [Kuester et al. \(2016\)](#) reinforce the point in a model where income composition differs across the distribution: monetary policy redistributes because households at the bottom rely on labour income while those at the top draw on capital and financial income, reshaping the transmission of policy to consumption while leaving output largely unaffected. This literature establishes both that monetary policy has measurable distributional consequences and that capturing them demands departing from the representative agent—yet in each case incidence is driven by wealth, portfolio, or financial-income heterogeneity, and the households are attached to no automation technology. The representative-agent frameworks that do model robotic capital ([Lin and Weise, 2019](#); [Fueki and Maehashi, 2019](#); [Basso and Rachedi, 2025](#)) are, conversely, structurally unable to speak to incidence at all.

There is, therefore, an established literature on (i) monetary policy interacting with robotic capital,

(ii) heterogeneity among households, and (iii) heterogeneity combined with generic capital (Bilbiie et al., 2022). Skill differentiation alongside automation is not itself new: Leduc and Liu (2024) pair the two, but in a real model with a representative household, where neither a policy rate nor an asset market constraint is present. What no existing framework does is combine automation capital that substitutes for unskilled labour with households excluded from owning it, in a setting where monetary policy transmits. Even where skill-differentiated capital appears in such a setting (Dolado et al., 2021), it complements skilled labour rather than substituting for the constrained agent's. This is the gap the present paper aims to fill.

2.1 The source of the divergence from existing results

Since the aggregate result reported below runs opposite in sign to the established finding that automation flattens the inflation–activity relationship, it is worth being exact about where the difference lies, and in particular about where it does not. It does not lie in the timing convention on the automated stock. Fueki and Maehashi (2019) accumulate labour-substitute capital through a conventional law of motion with investment adjustment costs, so their robot stock is predetermined within the period of a shock exactly as it is here, and their production structure is the same nested constant-elasticity arrangement, with an outer nest over traditional capital and labour services and an inner nest over robots and labour. Nor does it lie in the timing convention in Basso and Rachedi (2025), whose channel is the erosion of workers' outside option in a search-and-matching environment: that mechanism is a property of the bargaining protocol and would survive a pre-determined stock unchanged. It operates on a different margin from the one studied here and the two results do not compete.

The difference lies in the object measured and the horizon over which it is measured. Fueki and Maehashi (2019) simulate their model and estimate the Phillips curve on the artificial data, reporting that the slope against the output gap falls as robotisation advances while the slope against real marginal cost is left essentially unchanged; their flattening is therefore a statement about the mapping from activity to marginal cost, averaged across shocks and across the sample. The object here is the ratio of impact responses to a single identified monetary innovation. The two are not in conflict, and their own results make this explicit: they report that under a monetary policy shock the initial responses of inflation and of marginal cost become more pronounced as robotisation advances, with the attenuation appearing only at later horizons. That impact effect is what this paper isolates.

A framework in which the automated margin does adjust within the period delivers the opposite sign, which isolates the assumption at issue. Fueki et al. (2023) embed an endogenous automation threshold in a New Keynesian model, so that the allocation of tasks between capital and labour is reset each period by arbitrage between the rental cost and the wage. They report impulse responses to a monetary shock with and without that margin active. When it is free, the threshold moves, firms substitute towards capital within the period, the rise in real wages is smaller and the rise in rental costs larger, the response of the output gap is magnified and the response of inflation dampened through the Phillips curve. That is the comparison made here, run in the opposite di-

rection: a margin that can absorb the shock flattens the trade-off, and one that cannot steepens it. Their experiment is a sustained easing rather than a single innovation and combines the automation margin with variable utilisation and nominal wage rigidity, so the comparison establishes the direction of the effect rather than its size.

3 The model

To address the main question of this paper, we use a Two-Agent New Keynesian model based on [Bilbiie \(2008\)](#) and [Fueki and Maehashi \(2019\)](#). Households are heterogeneous along two dimensions: the type of labour they supply, skilled or unskilled, and whether they participate in asset markets. Participation determines behaviour, since households with access to assets are Ricardian—i.e., they consume and save—while those without are hand-to-mouth—i.e., they consume all current income. Skill determines the wage faced, and it is the unskilled wage against which robotic capital competes. Ricardian households can save in traditional capital, robotic capital, and a state-contingent asset that spans the return to firm ownership. Crossing the two dimensions gives four household cells but only two consumption behaviours, so the model retains the two-agent structure while allowing the skill composition of the constrained group to be set independently of its size.

The supply side is composed of final-good firms, which operate in a perfectly competitive market and produce the final good by aggregating intermediate goods, and intermediate-good firms, which operate in monopolistically competitive markets and sell differentiated intermediate goods to the final-good firm.

The choice of a two-agent rather than a full heterogeneous-agent structure is deliberate. [Debortoli and Galí \(2021\)](#) show that a Two-Agent New Keynesian model with a constant share of constrained households and no within-type heterogeneity approximates the aggregate-output dynamics of a canonical HANK model well, because the channel through which heterogeneity shapes aggregate fluctuations is dominated by the consumption gap between constrained and unconstrained households rather than by dispersion within either group. [Kaplan et al. \(2018\)](#) reach a complementary conclusion from the transmission side, showing that a two-agent structure reproduces the split between the direct and indirect effects of monetary policy that characterises richer models. Both results bear directly on our object of study: the incidence we analyse operates between household types—Ricardian versus hand-to-mouth—which is precisely the margin these approximations preserve, and we make no claim about dispersion within either type, which a two-agent model cannot capture. What the structure buys is not a closed-form solution, since the predetermined stock of robotic capital breaks the three-equation reduction available in the labour-only model of [Bilbiie \(2008\)](#). What it does secure is a finite household side: it spares the model an evolving wealth distribution as an additional state, leaving a linear rational-expectations system that remains solvable and, across our automation regimes, interpretable.

3.1 Households

There is a continuum of households of unit mass, partitioned along two dimensions. A fraction μ supplies unskilled labour and the complementary fraction $1 - \mu$ supplies skilled labour; skill type is fixed, with no reallocation between the two over the horizons considered here. Within each skill group, a fraction of households holds no assets and consumes all current income each period: λ_S among the skilled and λ_U among the unskilled. The aggregate hand-to-mouth share is therefore $\lambda = (1 - \mu)\lambda_S + \mu\lambda_U$, and the four cells carry masses $(1 - \mu)(1 - \lambda_S)$, $\mu(1 - \lambda_U)$, $(1 - \mu)\lambda_S$ and $\mu\lambda_U$. We use “Ricardian” and “saver” interchangeably for unconstrained households, index skill types by $j \in \{S, U\}$, and mark hand-to-mouth quantities with a superscript h . All households share the same utility function with common preference parameters, save for the disutility weight ω_j , so that all inequality in the model arises from differences in asset access and in wages rather than from preference heterogeneity.

All household variables are stated at the individual level: $N_{U,t}^h$ is the labour supplied by a single unskilled hand-to-mouth household, and so on. The corresponding economy-wide quantity is obtained by integrating over the relevant mass, so that unskilled hand-to-mouth labour supply is $\mu\lambda_U N_{U,t}^h$. An economy-wide total is accordingly smaller than the individual quantity it aggregates whenever the cell in question is a strict subset of the population; this reflects the normalisation of population to unity and carries no implication about the scale of individual behaviour.

Keeping the two dimensions distinct matters for what the incidence results can be attributed to, and the structure imposes no relation between them. The adding-up condition leaves one of λ_S and λ_U free once λ and μ are set, so the correlation between skill and participation is governed by a single parameter rather than imposed by construction. The benchmark sets $\lambda_S = \lambda_U = \lambda$, under which participation is independent of skill and constrained households are distributed across skill types in population proportions. The one-to-one mapping is the opposite corner of the same parameter, recovered at $\lambda_S = 0$, $\lambda_U = 1$ and $\mu = \lambda$. The benchmark is conservative rather than neutral, since the micro evidence points to a positive association between skill and participation; Section 4.1 sets out that evidence and the sense in which it is only a proxy for the constraint imposed here.

What the mechanism requires is in any case weaker than correlation between the two dimensions. It requires that the factor for which robotic capital substitutes command a wage that falls disproportionately, and that some of the households supplying that factor be unable to smooth the movement. Independence delivers the second condition in population proportions rather than concentrating it, so the incidence operates through the wage differential across skill types rather than through the composition of the constrained group; a positive correlation would amplify the result without changing its source.

Nor is the partition doing hidden identification work. Appendix C.4.2 retains the skill-differentiated production structure and the predetermined robot stock while removing the hand-to-mouth constraint, so the contribution of asset-market exclusion is separated from that of the nesting structure directly rather than assumed. The aggregate steepening result survives that removal, and so does

the incidence on unskilled factor income: unskilled labour income still deepens across regimes when no household is constrained. What does not survive is the transmission of that incidence to consumption, which is attributed to the constraint accordingly.

3.1.1 Ricardian Households

Ricardian households have mass $1 - \lambda$. Of these, $(1 - \mu)(1 - \lambda_S)$ supply skilled labour and $\mu(1 - \lambda_U)$ supply unskilled labour, where μ is the unskilled share of the population and λ_S and λ_U are the hand-to-mouth shares within the skilled and unskilled groups respectively. We write $n_S \equiv (1 - \mu)(1 - \lambda_S)/(1 - \lambda)$ and $n_U \equiv \mu(1 - \lambda_U)/(1 - \lambda)$ for the shares of the two skill types within the Ricardian group, with $n_S + n_U = 1$.

The asset B_t is a full vector of state-contingent claims, traded among Ricardian households only, with R_t the risk-free rate implied by their prices. Markets are complete in consumption but not in leisure, so with preferences separable in the two the marginal utility of consumption is common to all households holding the menu, and hours alone differ with the wage received (Erceg et al., 2000). The group is therefore written below as a representative family of mass $1 - \lambda$ that pools income into a single budget constraint, a formulation observationally equivalent to the decentralisation. The claims do not extend across the participation margin: hand-to-mouth households are excluded from the menu, and it is that exclusion, rather than the structure within the Ricardian group, that carries the mechanism.

Investment in either capital stock is subject to a cost of changing the rate of accumulation, and the two investment goods are not priced identically. We adopt the investment adjustment cost of Christiano et al. (2005), under which the resource loss depends on the growth rate of the investment flow rather than on its level relative to the installed stock,

$$S_j \left(\frac{I_{j,t}}{I_{j,t-1}} \right) = \frac{\tau_j}{2} \left(\frac{I_{j,t}}{I_{j,t-1}} - 1 \right)^2, \quad j \in \{K, Z\}, \quad (1)$$

with $\tau_K, \tau_Z \geq 0$. The function satisfies $S_j(1) = S'_j(1) = 0$ and $S''_j(1) = \tau_j$, so the cost vanishes to first order at the deterministic steady state and binds only along the transition. Separately, let p_t^Z denote the price of robotic investment goods in terms of the final consumption good, with the traditional investment good taken as numeraire. The two prices are distinct objects: the same nominal expenditure does not purchase the same quantity of consumption and of robotic capital, and a shock to p_t^Z alters the terms on which automation can be acquired without disturbing the consumption margin directly. We treat p_t^Z as exogenous and stationary,

$$\log p_t^Z = (1 - \rho_p) \log \bar{p}^Z + \rho_p \log p_{t-1}^Z + \varepsilon_{p,t}, \quad \varepsilon_{p,t} \sim \mathcal{N}(0, \sigma_p^2), \quad (2)$$

with $\rho_p \in [0, 1)$. Because the paper is not concerned with growth, no deterministic trend is imposed and the relative price fluctuates around a constant mean $\bar{p}^Z$.

The process is specified here in full because it is the natural second experiment and because the estimation exercise this model is intended to support requires it, but the relative price is held at

its mean in every experiment reported below, so that the comparison across regimes isolates the response to monetary policy alone.

The representative Ricardian family solves, per member,

$$\begin{aligned}
\max \quad & E_t \sum_{i=0}^{\infty} \beta^i \left[\frac{(C_{t+i}^r)^{1-\gamma}}{1-\gamma} - n_S \omega_S \frac{(N_{S,t+i}^r)^{1+\varphi}}{1+\varphi} - n_U \omega_U \frac{(N_{U,t+i}^r)^{1+\varphi}}{1+\varphi} \right] \\
\text{s.t.} \quad & C_t^r + I_{K,t} + p_t^Z I_{Z,t} + \frac{B_{t+1}}{P_t R_t} \leq n_S \frac{W_{S,t}}{P_t} N_{S,t}^r + n_U \frac{W_{U,t}}{P_t} N_{U,t}^r + r_t^K K_t + r_t^Z Z_t + \frac{B_t}{P_t} + \frac{1}{1-\lambda} D_t \\
& K_{t+1} = (1 - \delta_k) K_t + \left[1 - S_K \left(\frac{I_{K,t}}{I_{K,t-1}} \right) \right] I_{K,t} \\
& Z_{t+1} = (1 - \delta_z) Z_t + \left[1 - S_Z \left(\frac{I_{Z,t}}{I_{Z,t-1}} \right) \right] I_{Z,t} \\
& 0 \leq N_{S,t}^r, N_{U,t}^r \leq 1
\end{aligned}$$

where C_t^r denotes real consumption per Ricardian member, $N_{S,t}^r$ and $N_{U,t}^r$ the labour supplied by a skilled and an unskilled Ricardian member respectively, K_t the stock of traditional capital and Z_t the stock of robotic capital held per Ricardian member, $I_{K,t}$ and $I_{Z,t}$ the corresponding real investment flows, B_t holdings of the state-contingent asset, r_t^K and r_t^Z the real rental rates of the two capital stocks, and D_t aggregate real dividends. On the parameter side, $\beta \in (0, 1)$ is the intertemporal discount factor, $\gamma > 0$ is the coefficient of relative risk aversion (with $1/\gamma$ the intertemporal elasticity of substitution), $\omega_S, \omega_U > 0$ govern the weight of labour disutility for each skill type, $\varphi > 0$ is the inverse of the Frisch elasticity of labour supply, $\delta_k, \delta_z \in (0, 1)$ are the depreciation rates of traditional and robotic capital, and τ_K, τ_Z govern the curvature of the two adjustment cost functions. The logarithmic case of [Bilbiie \(2008\)](#) is nested at $\gamma = 1$. Because income is pooled, consumption per member is common to the two skill types and only hours differ; the ratio ω_S/ω_U together with the relative wage governs that difference. The benchmark sets this ratio to unity, so that the whole of the steady-state difference in hours between skill types comes from the relative wage and none of it from preference heterogeneity; the normalisation is recorded with the others in [Appendix B.1](#).

Three features of the constraint set should be noted. Firm equity does not appear as a separately traded asset: equity and the state-contingent claims are both financial claims priced by the same stochastic discount factor, so the equity margin is redundant and only the income it pays survives ([Bilbiie, 2008](#)). That income is aggregate dividends D_t , which accrue entirely to the mass $1 - \lambda$ of Ricardian households, hence the per-member term $\frac{1}{1-\lambda} D_t$. The two capital stocks are not redundant in the same sense: they are physically distinct factors, they enter production at different points of the nest and they depreciate at different rates, so each carries its own accumulation equation and its own predetermined state. Finally, the adjustment costs are imposed symmetrically on the two stocks. Restricting them to robotic capital alone would make automation uniquely frictional by construction, and any divergence between the responses of K_t and Z_t would then reflect that

asymmetry rather than the position of the two factors in the production nest.

Let $\Lambda_t = (C_t^r)^{-\gamma}$ denote the marginal utility of consumption, and let $\Lambda_t Q_{K,t}$ and $\Lambda_t Q_{Z,t}$ be the multipliers on the two accumulation equations, so that $Q_{K,t}$ and $Q_{Z,t}$ are the shadow values of a unit of installed capital of each type measured in consumption goods. Maximisation yields, in addition to the two intratemporal conditions, a bond Euler equation, a pair of investment conditions and a pair of valuation conditions:

$$(C_t^r)^{-\gamma} = \beta R_t \mathbb{E}_t \left[(C_{t+1}^r)^{-\gamma} \frac{P_t}{P_{t+1}} \right], \quad (3)$$

$$1 = Q_{K,t} [1 - S_K(x_{K,t}) - S'_K(x_{K,t}) x_{K,t}] + \beta \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} Q_{K,t+1} S'_K(x_{K,t+1}) x_{K,t+1}^2 \right], \quad (4)$$

$$p_t^Z = Q_{Z,t} [1 - S_Z(x_{Z,t}) - S'_Z(x_{Z,t}) x_{Z,t}] + \beta \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} Q_{Z,t+1} S'_Z(x_{Z,t+1}) x_{Z,t+1}^2 \right], \quad (5)$$

$$Q_{K,t} = \beta \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} (r_{t+1}^K + (1 - \delta_k) Q_{K,t+1}) \right], \quad (6)$$

$$Q_{Z,t} = \beta \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} (r_{t+1}^Z + (1 - \delta_z) Q_{Z,t+1}) \right], \quad (7)$$

$$\omega_S (N_{S,t}^r)^\varphi (C_t^r)^\gamma = \frac{W_{S,t}}{P_t}, \quad (8)$$

$$\omega_U (N_{U,t}^r)^\varphi (C_t^r)^\gamma = \frac{W_{U,t}}{P_t}, \quad (9)$$

where $x_{j,t} \equiv I_{j,t}/I_{j,t-1}$ for $j \in \{K, Z\}$. Conditions (4) and (5) equate the effective purchase price of a unit of investment to its shadow value net of the marginal adjustment cost incurred today and the marginal cost saved tomorrow by having raised the base against which next period's investment growth is measured. Conditions (6) and (7) price the installed stock: a unit installed at t pays the rental at $t+1$ and leaves the undepreciated fraction, valued at next period's shadow price, available thereafter. The within-family population weights n_S and n_U cancel from (8) and (9), because each weights the disutility of a member's labour and that member's contribution to family income identically.

Relative to a specification without adjustment costs, the purchase price and the shadow value of installed capital no longer coincide outside the steady state. Setting $\tau_K = \tau_Z = 0$ collapses (4) and (5) to $Q_{K,t} = 1$ and $Q_{Z,t} = p_t^Z$, at which point (6) and (7) reduce to the familiar rental-plus-depreciation Euler equations and the three intertemporal margins can be equated directly into a single no-arbitrage condition. With $\tau_j > 0$ that reduction is unavailable: the wedge between p_t^Z and $Q_{Z,t}$ is precisely the marginal adjustment cost, and it is what prevents the robotic investment flow from jumping to its new optimum on impact. The portfolio is still priced by a common stochastic discount factor, so the bond return continues to discipline the expected returns on both capital stocks, but the return on each now includes a capital gain term in its own shadow price.

In the deterministic steady state $x_K = x_Z = 1$, so $S_j = S'_j = 0$ and the adjustment costs are inoperative. Conditions (4) and (5) then give $Q_K = 1$ and $Q_Z = \bar{p}^Z$, and substituting into (6) and (7) yields

$$\bar{r}^K = \frac{1}{\beta} - (1 - \delta_k), \quad \bar{r}^Z = \bar{p}^Z \left[\frac{1}{\beta} - (1 - \delta_z) \right]. \quad (10)$$

The rental rate on traditional capital is unaffected by the extension. The rental rate on robotic capital is scaled by the mean relative price: a permanently cheaper robotic investment good lowers the rental the stock must earn in equilibrium, and therefore shifts the robotic share of the cost of the composite input. The adjustment cost parameters do not enter (10) at all, so τ_K and τ_Z are disciplined by the dynamics alone and leave the steady-state calibration intact. The choice of $\bar{p}^Z$ is discussed in Section 4.1.

3.1.2 Hand-to-Mouth Households

Hand-to-mouth households have mass λ , of which $(1 - \mu)\lambda_S$ supply skilled labour and $\mu\lambda_U$ supply unskilled labour. They hold no assets and cannot insure, so there is no counterpart here to the Ricardian family: the two skill types solve separate static problems and their consumption levels move independently. For $j \in \{S, U\}$,

$$\begin{aligned} \max \quad & E_t \sum_{i=0}^{\infty} \beta^i \left[\frac{(C_{j,t+i}^h)^{1-\gamma}}{1-\gamma} - \omega_j \frac{(N_{j,t+i}^h)^{1+\varphi}}{1+\varphi} \right] \\ \text{s.t.} \quad & C_{j,t}^h = \frac{W_{j,t}}{P_t} N_{j,t}^h \\ & 0 \leq N_{j,t}^h \leq 1 \end{aligned}$$

where $C_{j,t}^h$ denotes real consumption of a hand-to-mouth household of skill type j , $N_{j,t}^h$ its labour supply, and $W_{j,t}/P_t$ the corresponding real wage. Since these households hold no savings instrument, their consumption equals their labour income each period and the budget constraint holds with equality. The intratemporal condition is

$$\omega_j (N_{j,t}^h)^\varphi (C_{j,t}^h)^\gamma = \frac{W_{j,t}}{P_t}. \quad (11)$$

Substituting the budget constraint into (11) gives $\omega_j (N_{j,t}^h)^{\varphi+\gamma} = (W_{j,t}/P_t)^{1-\gamma}$, so that hand-to-mouth hours and consumption are

$$N_{j,t}^h = \omega_j^{-\frac{1}{\varphi+\gamma}} \left(\frac{W_{j,t}}{P_t} \right)^{\frac{1-\gamma}{\varphi+\gamma}}, \quad C_{j,t}^h = \omega_j^{-\frac{1}{\varphi+\gamma}} \left(\frac{W_{j,t}}{P_t} \right)^{\frac{1+\varphi}{\varphi+\gamma}}, \quad (12)$$

or, log-linearised, $\hat{N}_{j,t}^h = \frac{1-\gamma}{\varphi+\gamma} \hat{w}_{j,t}$ and $\hat{C}_{j,t}^h = \frac{1+\varphi}{\varphi+\gamma} \hat{w}_{j,t}$. Consumption is tied to current labour income rather than pinned by an Euler equation, so the wage enters (12) with an exponent proportional to $1 - \gamma$ and the hours elasticity $\frac{1-\gamma}{\varphi+\gamma}$ changes sign at $\gamma = 1$. At that value the wage cancels from the hours schedule, income and substitution effects offset exactly, hours are constant at $N_{j,t}^h = \omega_j^{-1/(1+\varphi)}$ and the whole of the wage movement is absorbed by consumption; this is the King-Plosser-Rebelo (King et al., 1988) case used by Bilbiie (2008). Below unity the offset is incomplete in the direction of substitution and hours co-move with the wage; above unity the income effect dominates and hours move against it. The elasticity applies to both skill types, since the two cells differ only in the wage they face and in ω_j .

The four cells are indexed by two shares. Skill type determines which wage the household faces, and participation status determines whether a movement in that wage passes into consumption or into the intertemporal margin. The adding-up condition $(1 - \mu)\lambda_S + \mu\lambda_U = \lambda$ leaves one of the two shares free once λ and μ are set, so the correlation between the two dimensions is governed by a single parameter. Setting $\lambda_S = 0$ and $\lambda_U = 1$ with $\mu = \lambda$ collapses the four cells to two and recovers the one-to-one mapping in which skill and participation coincide.

Aggregate labour supplies follow by integrating over the four cells:

$$\bar{N}_{S,t} = (1 - \mu) \left[\lambda_S N_{S,t}^h + (1 - \lambda_S) N_{S,t}^r \right], \quad (13)$$

$$\bar{N}_{U,t} = \mu \left[\lambda_U N_{U,t}^h + (1 - \lambda_U) N_{U,t}^r \right], \quad (14)$$

and these are the quantities that enter the production function of Section 3.2.

3.2 Firms

Following standard New Keynesian structure, there are two types of firms: final-good firms and intermediate-good firms. Final-good firms operate in a perfectly competitive market, whereas each intermediate good is produced by a monopolist.

The distinctive feature of our model is that intermediate firms employ two differentiated types of labour, skilled and unskilled, alongside robotic capital, which acts as a substitute for unskilled labour in production.

3.2.1 Final-good firms

Final-good firms produce the final good by aggregating a continuum of intermediate goods using a CES technology:

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} di \right)^{\frac{\varepsilon_p}{\varepsilon_p - 1}}$$

where $\varepsilon_p > 1$ is the price elasticity of demand for intermediate goods. Cost minimisation by the final-good firm yields the demand for intermediate good i ,

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t,$$

with P_t the price index dual to the aggregator, derived in [Appendix A.4](#).

3.2.2 Intermediate-good firms

A monopolist produces intermediate good i with a Cobb–Douglas technology in traditional capital and a labour–robot bundle X_t . The upper nest of that bundle combines skilled labour with a composite input L_t , which itself aggregates robotic capital and unskilled labour:

$$\begin{aligned} Y_t &= A_t \bar{K}_t^\alpha X_t^{1-\alpha} \\ X_t &= \left[\theta_S \bar{N}_{S,t}^{\rho_X} + (1 - \theta_S) L_t^{\rho_X} \right]^{\frac{1}{\rho_X}} \\ L_t &= \left[\theta_Z \bar{Z}_t^{\rho_L} + (1 - \theta_Z) \bar{N}_{U,t}^{\rho_L} \right]^{\frac{1}{\rho_L}} \end{aligned}$$

where A_t is total factor productivity, which follows a stationary autoregressive process, and $\alpha \in (0, 1)$ is the share of traditional capital in the outer layer. The substitution parameters map into elasticities as $\rho_X = (\sigma_X - 1)/\sigma_X$ and $\rho_L = (\sigma_L - 1)/\sigma_L$, so that $\sigma_X > 0$ is the elasticity of substitution between skilled labour and the composite input and $\sigma_L > 0$ that between robotic capital and unskilled labour. The share parameters are $\theta_S \in (0, 1)$ for skilled labour in the upper nest and $\theta_Z \in (0, 1)$ for robotic capital in the lower nest. The arguments of the technology are the population-weighted factor supplies defined above: $\bar{K}_t$ and $\bar{Z}_t$ from the Ricardian block, and $\bar{N}_{S,t}$ and $\bar{N}_{U,t}$ from [\(13\)](#) and [\(14\)](#), which already integrate over the constrained and unconstrained cells of each skill type.

Denote real factor prices by r_t^K , $w_{S,t} \equiv W_{S,t}/P_t$, $w_{U,t} \equiv W_{U,t}/P_t$ and r_t^Z . Both nests are homogeneous of degree one in their arguments, the outer layer is Cobb–Douglas and A_t is Hicks-neutral,

so the technology as a whole is homogeneous of degree one and unit cost is independent of the scale at which the firm operates. Factor markets being competitive, every intermediate producer faces the same factor prices and the same A_t , so the unit cost MC_t is common across i , which is what permits the pricing problem below to be solved once for a representative resetting firm; Appendix A.5 sets out the argument. The fixed cost Φ introduced below is a constant in final-good units and shifts total but not marginal cost.

The relative price of robotic investment does not enter the firm's problem. The firm rents the installed stock $\bar{Z}_t$ at the rental r_t^Z and is indifferent to the terms on which that stock was acquired; p_t^Z reaches marginal cost only indirectly, through the effect of the household's valuation condition on the equilibrium rental. The same applies to the adjustment costs, which are borne by the household on the investment flow rather than by the firm on the input.

Capital is rented on an economy-wide market rather than held firm-specifically, which sidesteps the strategic complementarity in price setting that arises when resetting firms face an upward-sloping short-run marginal cost curve of their own. That channel flattens the Phillips curve and would work against the steepening reported here. It is a distinct margin from the one studied in this paper, and nothing in the mechanism of Section 4.4 depends on shutting it down, since the amplification of (29) operates on the aggregate factor mix rather than on the elasticity of an individual firm's marginal cost.

Cost minimisation proceeds from the inside out, and is carried out in Appendix A.6. In the lower nest, minimising $r_t^Z \bar{Z}_t + w_{U,t} \bar{N}_{U,t}$ subject to the aggregator for L_t yields the unit cost of the composite input,

$$P_{L,t} = \left[\theta_Z^{\sigma_L} (r_t^Z)^{1-\sigma_L} + (1 - \theta_Z)^{\sigma_L} w_{U,t}^{1-\sigma_L} \right]^{\frac{1}{1-\sigma_L}}. \quad (15)$$

In the upper nest, L_t enters at price $P_{L,t}$ alongside skilled labour at price $w_{S,t}$, giving the unit cost of the bundle

$$P_{X,t} = \left[\theta_S^{\sigma_X} w_{S,t}^{1-\sigma_X} + (1 - \theta_S)^{\sigma_X} P_{L,t}^{1-\sigma_X} \right]^{\frac{1}{1-\sigma_X}}, \quad (16)$$

and the outer Cobb–Douglas layer combines traditional capital at price r_t^K with the bundle at price $P_{X,t}$, so that real marginal cost is

$$MC_t = \frac{1}{A_t} \left(\frac{r_t^K}{\alpha} \right)^\alpha \left(\frac{P_{X,t}}{1-\alpha} \right)^{1-\alpha}, \quad (17)$$

with the associated demands $r_t^K \bar{K}_t = \alpha MC_t Y_t$ and $P_{X,t} X_t = (1 - \alpha) MC_t Y_t$. The conditional factor demands implied by each nest are reported in Appendix A.6; in each, a factor is demanded in proportion to its nest aggregate and falls in its own price relative to the unit cost of that nest, with elasticity σ_L in the lower nest and σ_X in the upper. Equations (15), (16) and (17) give real marginal cost in closed form: the robot rental and the unskilled wage enter through $P_{L,t}$, which combines with the skilled wage in $P_{X,t}$, which in turn combines with the rental on traditional capital.

Following Bilbiie (2008), we introduce a fixed cost $\Phi > 0$, expressed in units of the final good,

that normalises steady-state profits to zero; since dividends accrue to Ricardian households alone, a positive steady-state profit would endow savers with a permanent income stream and displace the consumption ratio against which the incidence results below are read.

$$D_t(i) = \frac{P_t(i)}{P_t} Y_t(i) - MC_t Y_t(i) - \Phi, \quad (18)$$

which aggregates over the symmetric continuum to $D_t = (1 - MC_t) Y_t - \Phi$. Because Φ is a constant in final-good units it does not enter cost minimisation and leaves (17) unaffected. Imposing $D^{ss} = 0$ at the steady-state markup pins the fixed cost to the steady-state operating surplus,

$$\Phi = (1 - MC^{ss}) Y^{ss} = \frac{1}{\varepsilon_p} Y^{ss}. \quad (19)$$

Steady-state profits are therefore zero by construction, but profits vary out of steady state, and that variation accrues entirely to Ricardian households through the term $\frac{1}{1-\lambda} D_t$. The normalisation removes profit income as a source of steady-state dispersion between household types; the wage differential between skill types and the net return on the two capital stocks, which accrues to Ricardian households alone, survive it.

Prices are set following a Calvo mechanism. Each period a fraction $1 - \theta_p$ of intermediate-good firms resets its price optimally, while the remaining fraction θ_p indexes to past inflation according to $P_{t+s}(i) = \Pi_{t-1,t+s-1}^{\zeta_p} P_t(i)$, where $\Pi_{t-1,t+s-1} = P_{t+s-1}/P_{t-1}$ is cumulative gross inflation and $\zeta_p \in [0, 1]$ the degree of indexation. A firm resetting in period t chooses its price to maximise the expected discounted value of future operating profits, discounted by the ratio of Ricardian marginal utilities since firms are owned by Ricardian households alone; the fixed cost is independent of the price and is therefore absent from the first-order condition. Because real marginal cost is common across firms, every resetting firm chooses the same price, and under flexible prices that price collapses to the constant-markup benchmark $\frac{\varepsilon_p}{\varepsilon_p - 1} MC_t$ that underlies (19).

Appendix A.7 states the reset condition and the price index and carries out the log-linearisation around the zero-inflation steady state, which yields the New Keynesian Phillips curve

$$\hat{\pi}_t = \frac{\beta}{1 + \beta\zeta_p} \mathbb{E}_t \hat{\pi}_{t+1} + \frac{\zeta_p}{1 + \beta\zeta_p} \hat{\pi}_{t-1} + \frac{(1 - \theta_p)(1 - \beta\theta_p)}{\theta_p(1 + \beta\zeta_p)} \widehat{mc}_t, \quad (20)$$

which reduces to the purely forward-looking form $\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_p \widehat{mc}_t$ with $\kappa_p = (1 - \theta_p)(1 - \beta\theta_p)/\theta_p$ when $\zeta_p = 0$. We set $\zeta_p = 0$ throughout and retain the hybrid form here only for generality.

Log-linearising (17), (16) and (15) gives the marginal cost that enters (20),

$$\widehat{mc}_t = -\hat{A}_t + \alpha \hat{r}_t^K + (1 - \alpha) \left\{ s_S \hat{w}_{S,t} + (1 - s_S) [s_Z \hat{r}_t^Z + (1 - s_Z) \hat{w}_{U,t}] \right\}, \quad (21)$$

where

$$s_S \equiv \frac{w_S^{ss} \bar{N}_S^{ss}}{w_S^{ss} \bar{N}_S^{ss} + P_L^{ss} L^{ss}}, \quad s_Z \equiv \frac{r^{Z,ss} \bar{Z}^{ss}}{r^{Z,ss} \bar{Z}^{ss} + w_U^{ss} \bar{N}_U^{ss}}$$

are the steady-state cost share of skilled labour within the upper nest and the cost share of robotic capital within the lower nest. Both are nest-level shares and are distinct from the robot income share of output, Ω_Z , which indexes the automation regimes in Section 4.2. Each factor price enters marginal cost weighted by the product of the cost shares along the path from its nest to the top of the technology, so the robot rental and the unskilled wage reach the Phillips curve through the weight $(1 - \alpha)(1 - s_S)$.

The mean relative price of robotic investment enters s_Z through the steady-state rental, $\bar{r}^Z = \bar{p}^Z [1/\beta - (1 - \delta_z)]$, so $\bar{p}^Z$ joins θ_Z and σ_L as a parameter that reaches the Phillips curve only through the robotic cost share. A cheaper robotic investment good lowers the rental the stock must earn in equilibrium and therefore lowers s_Z at given quantities, while a higher θ_Z or a higher σ_L can raise it; the three cannot be identified separately from the slope of the inflation–activity relation alone. The adjustment cost parameters, by contrast, do not appear in (21) at all: they are inoperative at the steady state and so leave every cost share unchanged, affecting the Phillips curve only through the path the rental itself follows. This separation is used in Section 4.1, where $\bar{p}^Z$ is disciplined alongside the technology parameters and τ_K and τ_Z are disciplined by the dynamics.

Total factor productivity follows

$$\hat{A}_t = \rho_A \hat{A}_{t-1} + \varepsilon_t^A, \quad (22)$$

with $\varepsilon_t^A \sim \mathcal{N}(0, \sigma_A^2)$.

3.3 Aggregation and market clearing

Household variables are individual-level quantities within each cell, in the sense of Section 3.1. Aggregation therefore weights each cell by its population mass. Consumption aggregates as

$$C_t = (1 - \lambda) C_t^r + (1 - \mu) \lambda_S C_{S,t}^h + \mu \lambda_U C_{U,t}^h, \quad (23)$$

the labour aggregates $\bar{N}_{S,t}$ and $\bar{N}_{U,t}$ are those of (13) and (14), and the two capital stocks enter the technology as $\bar{K}_t = (1 - \lambda) K_t$ and $\bar{Z}_t = (1 - \lambda) Z_t$, since only Ricardian households accumulate them. Economy-wide investment in the two stocks is likewise $(1 - \lambda) I_{K,t}$ and $(1 - \lambda) I_{Z,t}$, the latter absorbing $(1 - \lambda) p_t^Z I_{Z,t}$ units of the final good. The state-contingent claims are in zero net supply within the Ricardian group, so $B_t = 0$ in equilibrium and the bond market clears without contributing a separate condition.

The goods market clears with the fixed cost entering as a genuine use of the final good and robotic investment valued at its relative price,

$$Y_t = C_t + (1 - \lambda)(I_{K,t} + p_t^Z I_{Z,t}) + \Phi. \quad (24)$$

The relative price therefore operates as a linear transformation technology between final goods and robotic capital, the economy surrendering p_t^Z units of output per unit of robotic capital installed. This is the reciprocal of the investment-specific efficiency term μ_Z of Fueki and Maehashi

(2019), who place the same object in the accumulation equation rather than in the constraint; the two formulations are equivalent up to that placement. The presence of Φ in (24) is what makes the constraint consistent with the zero-profit normalisation (19), and Appendix A.8 sets out the accounting.

The population shares are carried explicitly on the factor supplies rather than absorbed into θ_S and θ_Z . This is not a normalisation: it is the channel through which λ , μ , λ_S and λ_U enter the deterministic steady state, since the two labour types are distinct inputs earning distinct wages and the composition of the population therefore changes the effective supply of each.

The log-linear counterparts of (23) and (24) are

$$\hat{C}_t = \chi_r \hat{C}_t^r + \chi_{hS} \hat{C}_{S,t}^h + \chi_{hU} \hat{C}_{U,t}^h, \quad (25)$$

$$\hat{Y}_t = s_C \hat{C}_t + s_{IK} \hat{I}_{K,t} + s_{IZ} (\hat{I}_{Z,t} + \hat{p}_t^Z), \quad (26)$$

where

$$\begin{aligned} \chi_r &\equiv \frac{(1-\lambda)C_r^{ss}}{C^{ss}}, & \chi_{hS} &\equiv \frac{(1-\mu)\lambda_S C_S^{h,ss}}{C^{ss}}, & \chi_{hU} &\equiv \frac{\mu\lambda_U C_U^{h,ss}}{C^{ss}}, & s_C &\equiv \frac{C^{ss}}{Y^{ss}}, \\ s_{IK} &\equiv \frac{(1-\lambda)I_K^{ss}}{Y^{ss}}, & s_{IZ} &\equiv \frac{(1-\lambda)\bar{p}^Z I_Z^{ss}}{Y^{ss}}. \end{aligned}$$

By construction $\chi_r + \chi_{hS} + \chi_{hU} = 1$, whereas $s_C + s_{IK} + s_{IZ} = 1 - 1/\varepsilon_p < 1$, the remainder being the fixed cost. Since Φ is a constant in final-good units it has no deviation from steady state and so contributes no term to (26), while its presence in (24) is what makes the three expenditure shares sum to less than unity. The robotic expenditure share is valued at the mean relative price, so a shock to p_t^Z moves output on impact through the expenditure side even before the investment flow itself responds.

3.4 Monetary policy

The monetary authority sets the nominal interest rate with an instrument rule that responds to relevant macroeconomic indicators with partial adjustment, following Bilbiie (2008):

$$R_t = \mathcal{R} \left[R_{t-1}, R_t^*, E_t \frac{P_{t+1}}{P_t}, \frac{Y_t}{Y_t^*} \right] \quad (27)$$

where star variables denote the natural values of the corresponding variables, that is, their values under fully flexible prices, and $\mathcal{R}$ is an arbitrary function. The flexible-price benchmark Y_t^* is the level of output that would obtain in an otherwise identical economy in which $\theta_p = 0$, so that $MC_t = (\varepsilon_p - 1)/\varepsilon_p$ at all dates; the output gap is $\hat{x}_t \equiv \hat{Y}_t - \hat{Y}_t^*$. The elasticities of the function are

denoted by ϕ^* , ϕ_π and ϕ_x , and $\rho_R \in [0, 1)$ is the weight on the lagged rate. Adopting a Taylor rule with interest-rate smoothing, the log-linearisation of (27) is

$$\hat{R}_t = \rho_R \hat{R}_{t-1} + (1 - \rho_R) [\phi^* \hat{R}_t^* + \phi_\pi \hat{\pi}_t + \phi_x \hat{x}_t] + \varepsilon_t^R \quad (28)$$

where $\varepsilon_t^R \sim \mathcal{N}(0, \sigma_R^2)$ is a stochastic monetary shock. Throughout the quantitative analysis we set $\phi^* = 0$, so that the natural rate does not enter the rule directly; the term is retained in (28) because it is what nests the specification of Bilbiie (2008). The lagged rate enters undiscounted by $(1 - \rho_R)$, so the rule is a standard partial-adjustment specification rather than smoothing applied to the whole right-hand side, and $\hat{R}_{t-1}$ joins the state vector as a new predetermined variable alongside the lagged investment flows of Section 3.1. Note that a monetary shock leaves the flexible-price allocation unchanged, so that $\hat{Y}_t^* = 0$ and $\hat{x}_t = \hat{Y}_t$ in the experiments reported below; the distinction between the two would matter for a technology shock.

4 Results

This section turns the model into a quantitative statement in two stages. The first is preparatory: the calibration fixes the parameters common to every economy considered, and the experimental design sets out the grid of automation regimes, the sense in which those economies are comparable, and the normalisation under which they are compared. The second reports the experiment itself. The aggregate responses establish what a common monetary tightening does to activity and inflation as automation rises, the mechanism subsection traces that result to the predeterminacy of the robot stock and its weight in marginal cost, and the incidence subsection asks which households bear the adjustment. Three shorter subsections then take up the savers' side of the same mechanism, the shape of the transition beyond impact, and the sensitivity of the comparison to the normalisation adopted.

4.1 Calibration

Table 1: Calibration summary

β	γ	φ	ω	λ	μ	α	σ_X	σ_L	δ_k	δ_z	$\bar{p}^Z$
0.99	0.75	1.610	9	0.24	0.24	0.30	0.90	10	0.025	0.04	1
ε_p	θ_p	ζ_p	ϕ_π	ϕ_x	ρ_R	$\tau_K = \tau_Z$	ρ_p	σ_p	ρ_A	σ_{ε_A}	σ_{ε_R}
6	0.75	0	1.5	0.125	0.8	6.962	0.90	0.01	0.90	0.01	0.01

Table 1 reports the full set of parameter values used in the quantitative analysis. The choice of each parameter is discussed below, distinguishing values taken from the literature from those chosen in this work.

- **Preferences.** The discount factor follows the standard quarterly convention. Relative risk aversion is set below unity. The value is not estimated here, and the qualitative behaviour of the model depends on γ only through whether it lies above or below unity: at $\gamma = 1$ the hand-to-mouth employment margin is inactive, for $\gamma < 1$ unskilled hours fall when the unskilled wage falls, and for $\gamma > 1$ they rise. A contractionary shock that raised unskilled employment would be difficult to reconcile with the evidence discussed in Section 2, and $\gamma < 1$ is adopted on that ground; Appendix C.3 reports the alternative $\gamma = 1.25$, chosen to sit the same distance above the logarithmic case $\gamma = 1$ as the benchmark sits below it, rather than a value drawn from an external estimate. The inverse Frisch elasticity is the reciprocal of the estimate of Keane and Neal (2025). The labour disutility scale is a free level parameter, not externally targeted.¹
- **Households.** The hand-to-mouth share is treated as an exogenous primitive, calibrated to the euro-area estimate reported in Ampudia et al. (2018), who apply the wealth-based definition of Kaplan et al. (2014) to the second wave of the Household Finance and Consumption Survey and obtain a combined poor- and wealthy-hand-to-mouth share of $\lambda = 0.24$, in preference to the US-based figure of Kaplan et al. (2014) used in earlier drafts of this paper, on the ground that the euro-area estimate applies the same wealth-based classification to a population directly comparable to the one modelled here. The population share of unskilled households, μ , is calibrated to the routine employment share for Germany reported in Goos et al. (2014) and cross-checked against the later estimate of Oesch and Piccitto (2019), both of which place this share in a range of 0.20 to 0.30, with $\mu = 0.24$ adopted as the baseline.² The published statistical tables of the Household Finance and Consumption Survey do not cross-tabulate hand-to-mouth status with educational attainment, and microdata access to construct this joint distribution directly was not obtained; skill type and hand-to-mouth status are accordingly assumed independent, so that $\lambda_S = \lambda_U = \lambda$ and the skilled and unskilled shares of the Ricardian population reduce to $n_S = 1 - \mu$ and $n_U = \mu$. The micro evidence points the other way. Mankiw and Zeldes (1991) document that stock market participation rises steeply with the education of the household head and that the gradient survives conditioning on income: within the third quartile of average labour income, participation is under a fifth among heads without a high-school degree and above one half among college graduates, which they attribute to fixed information costs falling with education. Vissing-Jorgensen (2002) confirms that participation rises with education and provides direct evidence for the fixed-cost interpretation of non-participation itself, estimating that a

¹Preferences of the Greenwood–Hercowitz–Huffman form would deliver procyclical constrained hours at any value of relative risk aversion, removing the dependence of the sign on this parameter. They are not adopted here precisely because they do so by eliminating the wealth effect on labour supply, which is the margin through which the asset market constraint operates in this model. The dependence is a property of the mechanism rather than an inconvenience to be assumed away, and Appendix C.3 reports the model at a conventional value of relative risk aversion, where the qualitative structure of the distributional result survives and only the sign of constrained hours changes.

²The share of constrained households and the share of unskilled workers take the same value in the calibration. This is a coincidence of two independent sources, the euro-area household finance evidence and the routine employment share, and no restriction linking the two dimensions is imposed; the independence assumption is stated in Section 3.1.

per-period cost of around fifty dollars rationalises the choices of half of all non-participants; she reads the education effect, however, as reflecting a correlation between education and preferences rather than between education and participation costs. [Kaplan et al. \(2014\)](#) report an education gradient across the three groups they identify, with unconstrained households averaging about one more year of schooling than the wealthy hand-to-mouth and the latter about one more year than the poor hand-to-mouth. One qualification applies to all three. The participation literature measures non-participation in equity markets, whereas the constraint imposed here excludes households from all assets including the riskless bond; the former is the conventional microfoundation for limited asset market participation but is a proxy for the latter rather than a measurement of it, which is why the aggregate share is calibrated to a direct measure of hand-to-mouth status and only the within-group split is left at the independence benchmark. Setting $\lambda_U = \lambda_S$ therefore places fewer constrained households in the unskilled group than these gradients suggest, and the benchmark is conservative in the operative sense: any disproportionate unskilled hand-to-mouth incidence reported below is attributable to the model's structural mechanism rather than to an assumed empirical correlation between skill and financial constraint.

- **Technology.** The capital share in the outer nest is set to the conventional value used throughout quarterly business-cycle models, and traditional capital depreciates at the standard quarterly rate. The upper-nest elasticity of substitution is an author's calibration choice, differing from the value implied by [Perez-Laborda and Perez-Sebastian \(2020\)](#) under a reading of Z as regular (rather than robotic) capital, a reading not adopted here; the lower-nest elasticity is an author's choice approximating near-perfect substitutability between robots and unskilled labour, in the spirit of the task-based automation framework of [Acemoglu and Restrepo \(2020\)](#), rather than a fitted or literature-sourced value; robot depreciation is set above the rate assumed for traditional capital, at the upper end of the range over which [Anne et al. \(2022\)](#) reconstruct robot stocks by perpetual inventory. The implicit rates they compute from the published stocks are lower, but those measure the physical withdrawal of units under an assumed service life rather than the loss of productive value of an installed machine, and the service life in question is roughly double the one adopted in German and American tax depreciation schedules for the same equipment. Since the relative price of robotic investment goods is stationary in this model, obsolescence has no other place to enter, and the depreciation rate is read as the economic rate accordingly. The parameter is not load-bearing: because regimes are indexed by the robot income share, the cost share governing the mechanism is pinned by that target at any depreciation rate, and what the choice affects is the implied stock and the share of output absorbed by robot investment.
- **Investment dynamics.** The investment adjustment cost parameter, common to traditional and robotic capital under $\tau_K = \tau_Z$, follows the posterior mean estimated for the euro area in [Smets and Wouters \(2003\)](#), in preference to the value estimated for the United States in [Christiano et al. \(2005\)](#), both for geographic relevance and because the two studies share a common full-system Bayesian estimation approach, avoiding a mismatch between minimum-distance and likelihood-based estimates of the same structural parameter. The steady-state relative price of robotic investment goods, $\bar{p}^Z$, is normalised to one: since s_Z is a sufficient statistic for the transmission channel, no calibration target for the level of $\bar{p}^Z$ is required. No estimate of the persistence and volatility of the relative-price process is directly transportable to this specification, so ρ_p and σ_p are set equal to the values adopted for the technology process

below, ρ_A and σ_{ε_A} .

- **Nominal rigidities.** The elasticity of substitution across goods implies a standard New Keynesian steady-state markup; the Calvo parameter corresponds to a standard average price duration in the literature; price indexation is set to zero, a modelling choice yielding a purely forward-looking Phillips curve.
- **Monetary policy.** The rule nests [Bilbiie \(2008\)](#) as a special case; the contemporaneous Taylor coefficients follow standard values in the literature, and the weight on the natural rate is set to zero. The smoothing coefficient is set at $\rho_R = 0.8$ rather than at the posterior mean of 0.956 estimated for the euro area in [Smets and Wouters \(2003\)](#), since that value is estimated jointly with habit formation and a rule that also responds to the change in inflation and the output gap, features absent here.
- **Shocks.** TFP persistence and innovation standard deviations follow standard values in the NK/RBC literature.
- **Targets.** The skill premium target is an author's choice, selected as an empirically realistic value with no specific literature target; the robot income share is swept across a range of target values.

Two of these choices together determine the strength of monetary transmission in the model, and it is worth separating them from the mechanism. Relative risk aversion below unity implies an intertemporal elasticity of substitution of 1.33, and the degree of interest rate smoothing makes the real rate path following a given impact response persistent; the two compound, so a twenty-five basis point impact response of the policy rate produces an impact contraction in output larger than the empirical literature on monetary transmission would support, and larger than the near-zero impact response found in identified vector autoregressions. The level of transmission in this model is therefore not a quantity to be read off against the data. It is also not the object of any result reported below. Every statistic in what follows is either a ratio of two responses within a regime or a gradient across regimes, and both are invariant to a common rescaling of transmission strength: multiplying every response by a constant leaves the sacrifice ratio, the amplification factor and the excess burdens unchanged. What the calibration must deliver is the correct comparative statics in the robot income share, and the parameters that govern those are the cost shares, which are disciplined by the steady-state targets.

4.2 Quantitative analysis

We solve the model at first order and compute impulse responses to a contractionary monetary policy shock. Regimes are indexed by the robot income share of output, $\Omega_Z \equiv r^Z \bar{Z}/Y$, rather than by the CES share parameter θ_Z . The reason is that θ_Z is not a transportable measure of automation intensity: across the range of scenarios considered here it varies only within a narrow interval, while the robot income share it delivers varies by a factor of seven. Indexing by the income share

also has the advantage that the resulting economies can be assessed directly against observable aggregates.

For each regime, θ_Z and θ_S are solved jointly, inside the steady state, so as to satisfy two calibration targets simultaneously: the robot income share takes its assigned regime value, and the steady-state skill premium equals 1.7 in every regime. All remaining parameters are held fixed. Holding the skill premium fixed while raising the robot income share requires both share parameters of the technology to move, so the regimes differ in a pair of steady-state cost shares rather than in one: the robot share of the lower nest rises and the skilled share of the upper nest falls. The decomposition of marginal cost nonetheless isolates the automation margin. The weight of the robot rental in real marginal cost equals the robot income share divided by steady-state marginal cost, and is therefore fixed by the calibration target whatever the two share parameters do; the weight of traditional capital is the outer-nest share and does not move; and the residual weight divides between the two wages in a ratio that the skill premium target holds constant to within two per cent across the grid. The movement in the skilled cost share accommodates the rising robot weight rather than constituting an independent source of variation, and the regimes are comparable in that sense.

Impulse responses are normalised so that the impact response of the nominal policy rate equals twenty-five basis points in every regime. This compares regimes at a common policy action rather than at a common structural innovation, which is the object of interest here: the question is how a given tightening is transmitted in a more automated economy, not how a given disturbance to the rule propagates. The normalisation is not innocuous. Delivering the same impact rate response requires a structural innovation twenty-eight percent larger at $\Omega_Z = 0.35$ than at $\Omega_Z = 0.05$, because inflation responds more strongly in the more automated economies and the systematic component of the rule partly undoes the shock. Reporting at a common innovation instead would attribute part of that endogenous feedback to the transmission mechanism.

None of the regimes on the grid should be read as a calibrated economy. The robot income shares considered are not intended to match the share implied by the observed robot stock in any particular economy, and the highest of them implies an aggregate labour share below any that has been observed. The grid is a comparative-statics device rather than a menu of economies, and the object of interest is accordingly the gradient of the responses across regimes rather than the level of any response at a particular point on it. Read this way the exercise asks in which direction continued automation moves the transmission of monetary policy, and the highest regimes serve to make that direction legible at a scale where the mechanism is unambiguous, not to describe an economy that anyone expects to observe.

4.3 Aggregate responses

Table 2 reports the impact responses. Two aggregate patterns move in opposite directions. The contraction in output becomes progressively weaker as automation deepens, while the contraction

in inflation becomes progressively stronger. Monetary policy therefore becomes less effective at moving real activity and more effective at moving prices as the economy automates.

Table 2: Impact responses to a contractionary monetary policy shock, by robot income share.

Robot income share Ω_Z	0.05	0.15	0.25	0.35
Policy rate, $\hat{R}^n$	0.0025	0.0025	0.0025	0.0025
Output, $\hat{y}$	-0.0136	-0.0126	-0.0116	-0.0107
Inflation, $\hat{\pi}$	-0.0081	-0.0092	-0.0109	-0.0141
Real marginal cost, $\hat{m}c$	-0.0582	-0.0674	-0.0828	-0.1129
Aggregate consumption, $\hat{C}$	-0.0215	-0.0225	-0.0240	-0.0266
Ricardian consumption, $\hat{C}^r$	-0.0117	-0.0117	-0.0117	-0.0117
Skilled HtM consumption, $\hat{C}_S^h$	-0.0560	-0.0618	-0.0722	-0.0934
Unskilled HtM consumption, $\hat{C}_U^h$	-0.0642	-0.0787	-0.0965	-0.1289
Skilled wage, $\hat{w}_S$	-0.0506	-0.0559	-0.0653	-0.0845
Unskilled wage, $\hat{w}_U$	-0.0581	-0.0711	-0.0873	-0.1166
Skilled Ricardian hours, $\hat{N}_S^r$	-0.0260	-0.0293	-0.0351	-0.0470
Unskilled Ricardian hours, $\hat{N}_U^r$	-0.0306	-0.0387	-0.0487	-0.0669
Skilled HtM hours, $\hat{N}_S^h$	-0.0054	-0.0059	-0.0069	-0.0089
Unskilled HtM hours, $\hat{N}_U^h$	-0.0062	-0.0075	-0.0092	-0.0123
Robot rental rate, $\hat{r}^Z$	-0.0604	-0.0741	-0.0909	-0.1215
Capital rental rate, $\hat{r}^K$	-0.0718	-0.0800	-0.0944	-0.1237
Robot investment, $\hat{i}_Z$	-0.0019	-0.0018	-0.0018	-0.0017
Capital investment, $\hat{i}_K$	-0.0018	-0.0017	-0.0017	-0.0016

The divergence is best expressed as a trade-off. Table 3 reports the sacrifice ratio, which throughout this paper denotes the output cost of a unit of disinflation on impact, $\hat{y}/\hat{\pi}$, rather than the conventional cumulative output loss per point of permanent disinflation.³ It falls by thirty-seven percent over the range of robot income shares from five to twenty-five percent, and by rather more than half across the full grid. This is the central aggregate result, and it is reported as a ratio deliberately: both terms carry the same normalisation factor, so the trade-off is invariant to the choice made in Section 4.2 in a way that the individual responses are not.

³The two measures move together here because the response is heavily front-loaded: without interest rate smoothing, ninety percent of the cumulative output loss falls in the impact quarter at the lowest robot share. Appendix C.5 reports both, and the cumulative measure declines by thirty-six percent over the range for which the impact measure declines by forty-one. Nothing in what follows turns on the choice between them.

Table 3: Inflation–output trade-off on impact, and the amplification of composite-input movements onto unskilled hours.

Robot income share Ω_Z	0.05	0.15	0.25	0.35
Disinflation per unit of output, $\hat{\pi}/\hat{y}$	0.596	0.730	0.941	1.308
Sacrifice ratio, $\hat{y}/\hat{\pi}$	1.678	1.369	1.062	0.764
Amplification, $1/(1 - s_Z)$	1.75	3.75	6.95	12.86
Implied robot cost share, s_Z	0.427	0.733	0.856	0.922
Implied skilled cost share, s_S	0.799	0.649	0.499	0.349
Weight of the robot rental in $\widehat{mc}$	0.060	0.180	0.300	0.420
Weight of the skilled wage in $\widehat{mc}$	0.560	0.455	0.350	0.245
Weight of the unskilled wage in $\widehat{mc}$	0.081	0.066	0.050	0.035

This result is often read as running against the existing literature, which concludes that automation flattens the inflation-activity relationship. As Section 2.1 argues, the relation to that literature depends on which framework is in view. Against Fueki and Maehashi (2019), whose automated stock is predetermined as it is here, there is no conflict: their flattening concerns the average mapping from activity to marginal cost across shocks and horizons, and they themselves report a strengthened impact response under a monetary innovation. Against frameworks in which the automated margin adjusts within the period, the divergence is real and is driven by exactly that margin.

The composition of the output response deserves a remark, since it is not obvious how output can contract less while aggregate consumption contracts more. The reconciliation is in the expenditure weights. As the robot income share rises, robot investment absorbs a growing share of output and the weight attached to consumption in the goods-market identity falls correspondingly. A deeper proportional consumption response therefore contributes less to output. The attenuation of the output response is an accounting consequence of the changing expenditure composition as much as a behavioural one, and should not be read as evidence that private demand becomes less sensitive to policy.

4.4 The mechanism

It is important to be precise about where the steepening originates. The structural slope of the New Keynesian Phillips curve depends only on the Calvo parameter and is invariant across regimes by construction. What changes is the elasticity of real marginal cost with respect to output, and the source of that change is the rigidity of the input mix. Predeterminacy leaves two margins free to move within the period, and they are distinct: the quantity of unskilled labour, which must absorb the whole movement in the lower-nest composite, and the price of robot services, which must clear a market in which the quantity cannot move at all. The first governs the incidence of the disinflation and the second governs its aggregate cost, and the same steady-state cost share

governs both. Taking them in turn makes clear why the two results of this paper are one.

Cost minimisation over the lower nest makes the composite input a constant-elasticity aggregate of robotic capital and unskilled labour, so its log-linearisation is the cost-share-weighted average of the two factor movements, $\hat{L}_t = s_Z \hat{Z}_t + (1 - s_Z) \hat{N}_{U,t}$, where s_Z is the steady-state cost share of robots within the nest. The robot stock is predetermined, so $\hat{Z}_t = 0$ on impact and the whole movement in the composite falls on hours,

$$\hat{N}_{U,t} = \frac{\hat{L}_t}{1 - s_Z}. \quad (29)$$

The step is an accounting one and nothing is claimed for it beyond what it makes visible, but what it makes visible is the object the rest of this section turns on. The factor $1/(1 - s_Z)$ is unbounded as $s_Z \rightarrow 1$ and equals unity when robots are absent from the nest, so the cost share of the predetermined input is what converts a movement in the composite into a movement in hours. It depends on no other parameter. In particular it is invariant to the elasticity of substitution between robots and unskilled labour, which enters the steady state only through s_Z itself and is absorbed once the robot income share is targeted; Appendix C confirms this numerically, the amplification factor being identical at every value of that elasticity considered. The same cost share appears again below in the unit cost of the nest, and that recurrence rather than (29) itself is what ties the incidence result to the aggregate one.

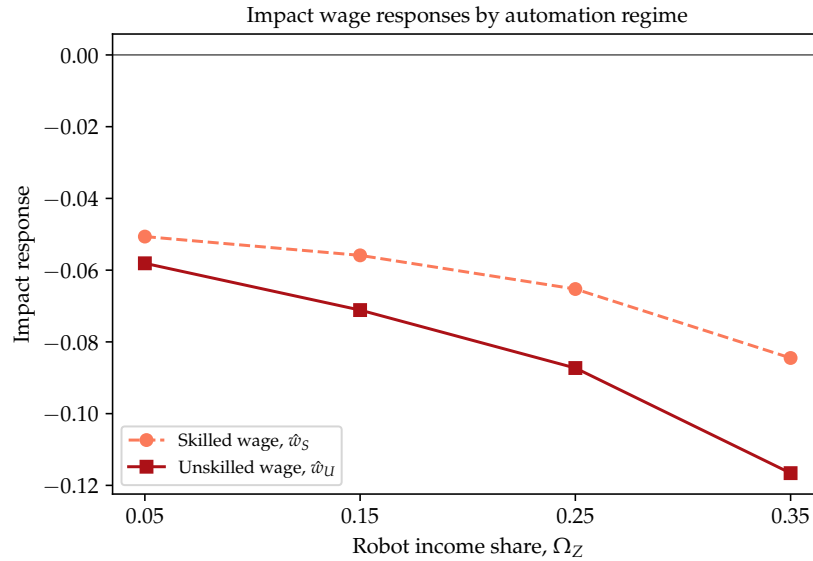


Figure 2: Wage divergence across regimes

Equation (29) concerns the quantity margin. Table 3 reports the amplification factor computed directly from the impact responses, and it rises by a factor of seven across the regimes considered:

as robots occupy a larger cost share of the nest, a given change in activity requires a proportionally larger movement in unskilled hours, and hence in the unskilled wage.

The aggregate result, however, runs through the price margin rather than this one, and the distinction matters because the two move in opposite directions in the cost weights. The unit cost of the lower-nest composite is the cost-share-weighted average of the two factor prices, $\hat{p}_L = s_Z \hat{r}^Z + (1 - s_Z) \hat{w}_U$, so the unskilled wage enters real marginal cost with weight $(1 - \alpha)(1 - s_S)(1 - s_Z)$, which falls to zero as the robot share rises, while the robot rental enters with a weight rising towards the full lower-nest share. Table 2 shows the consequence. At the lowest regime the composite cost response of -0.059 divides into -0.026 from the rental rate and -0.033 from the unskilled wage, and the two margins carry the adjustment jointly. At the highest regime the same decomposition is -0.104 from the rental rate and -0.008 from the wage: the unskilled wage is moving more than any other price in the table and contributing almost nothing to marginal cost. The steepening of the trade-off is therefore carried by the rental rate on a stock that cannot adjust, not by the wage on the input that adjusts instead. Both are consequences of the same predeterminacy, but they are different margins and they support different halves of the paper's result.

Hand-to-mouth labour supply then converts that movement into a larger wage movement still. Because these households consume their current labour income in full, the income effect holds their hours close to fixed, and the shock is absorbed on the wage margin rather than the quantity margin. The two amplifications compound, which is why the unskilled wage response deepens by nearly a factor of two across the grid while output deepens not at all, as Figure 2 shows.

The steepening as such is a generic consequence of factor predeterminacy and would arise with traditional capital in place of robots: raising the share of predetermined factors steepens the short-run marginal cost curve, and nothing in that statement requires the predetermined factor to be automation capital. What is specific to this model is the joint structure. The factor whose price carries the aggregate steepening sits in the same nest as the labour input whose hours carry the incidence, so a single cost share determines both how much cheaper disinflation becomes and who pays for it. Neither result is available from a model with one predetermined factor and homogeneous labour, and the aggregate and distributional findings of this section are two readings of one mechanism rather than two findings.

4.5 Incidence

Table 4: Excess burden of the contraction on constrained households.

Robot income share Ω_Z	0.05	0.15	0.25	0.35
Skilled HtM to Ricardian, $\hat{C}_S^h / \hat{C}^r$	4.76	5.17	5.91	7.39
Unskilled HtM to Ricardian, $\hat{C}_U^h / \hat{C}^r$	5.47	6.58	7.89	10.19
Unskilled to skilled HtM, $\hat{C}_U^h / \hat{C}_S^h$	1.147	1.272	1.336	1.379

Hand-to-mouth households consume their current labour income in full, so their consumption response is proportional to their wage response, with a factor of 1.106 at the benchmark calibration. This proportionality holds exactly at every regime and every horizon in the solution, and it means that the consumption gap between skilled and unskilled constrained households carries no information beyond the relative wage response. We report wages as the object of interest accordingly, and consumption as its image.

Table 4 reports the ratio of each constrained group’s consumption loss to the Ricardian loss. Two features stand out. The ratio is large at every level of automation, which is the standard result in models with limited asset market participation. More interestingly, it rises steeply with the robot income share, and it rises by more for unskilled than for skilled constrained households.

The two dimensions of the ratio can be read separately. The skilled column isolates the asset-market constraint: these households are constrained but sit in the upper nest, insulated from the lower-nest rigidity. The unskilled column adds position in the lower nest to the constraint. At the lowest regime the two are close: nest position accounts for about a sixth of the excess burden borne by unskilled constrained households, and the asset-market constraint for the remainder. At the highest regime they diverge substantially, and nest position accounts for close to a third.

These ratios are reported in preference to the underlying levels because a ratio of two responses within a regime is invariant to any common rescaling of the shock. Both terms carry the same scaling factor, so the profile across regimes is unaffected by the choice discussed in Section 4.2 and by the level of transmission discussed in Section 4.1. This is not true of the difference between two log deviations, which scales linearly with the shock and would therefore inherit the twenty-eight per cent variation in the scaling factor documented in Section 4.8. The choice of normalisation continues to matter for the levels reported in Table 2, and for the reading of Ricardian consumption in particular, but not for the comparisons drawn here.

One restriction of the household structure limits how far this decomposition can be taken. Ricardian consumption is pooled across skill types within the family, so skill enters the Ricardian block through hours but not through consumption. The model therefore delivers three distinct consumption responses rather than four. This is full insurance within the saver household rather than an omission, and it is appropriate to questions that do not turn on incomplete markets among unconstrained agents; what it costs is the ability to ask whether skilled and unskilled savers differ in their exposure to policy, a question this framework sets aside rather than answers.

4.6 Savers and the investment margin

Ricardian consumption is close to invariant across regimes, moving only in the third decimal place. This is not because savers are unaffected by the shock but because they are affected in an offsetting way: their factor income becomes markedly more exposed to monetary policy as automation deepens, since a growing share of it is robot rental income and the rental rate on a predetermined

stock absorbs the whole of the short-run adjustment. Consumption is nevertheless governed by the intertemporal condition, and the real rate path is what the normalisation holds fixed.

The margin through which savers absorb the income shock is robot investment, and the presence of adjustment costs restricts that margin severely. The impact response of robot investment is roughly a sixth of the output response, where in a frictionless specification it exceeds output by an order of magnitude. The reason is visible in the valuation condition: the rental rate on a predetermined stock spikes on impact and returns almost fully by the following quarter, so the expected path of future rentals barely moves, the shadow value of installed robots moves with it, and investment has little to respond to. The model carries no habit formation and no nominal wage rigidity, so the real block has little internal propagation and the disturbance is concentrated in the impact quarter. Robot investment is nevertheless the only variable in the model displaying a hump-shaped response, peaking one quarter after impact rather than on impact, as Figure 3 shows. The consequence for the exercise should be stated plainly: the robot stock is effectively immobile at every horizon here, not merely on impact, so the comparison across regimes is closer to a comparative static in the robot cost share than to a study of automation dynamics. That is a limitation of what the experiment can say about adjustment over time, and at the same time it is what makes the predeterminacy reading of the results exact rather than approximate. The restriction on this margin is also the reason Ricardian consumption responds at all: with investment less able to absorb the shock, more of it must pass through to current consumption.

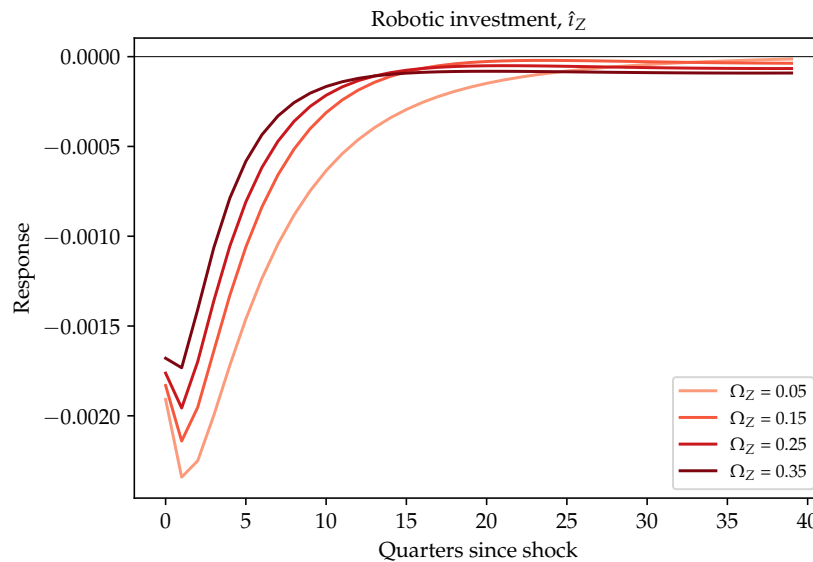


Figure 3: Robot investment dynamics

Income composition matters for savers even though it does not enter their consumption decision, and it moves in two opposing directions as automation deepens. Rental income becomes more exposed: the robot rental rate falls further the larger the robot income share, as Table 2 shows, and

it accounts for a growing share of saver income by construction. Dividend income moves the other way. Because prices are sticky, the contraction compresses real marginal cost by considerably more than output, so the markup widens and profits rise. Dividends are zero in steady state under the fixed-cost specification adopted here, so the model identifies the direction of this movement rather than its proportional size; what can be said is that it is positive, that it is larger the more automated the economy, and that it is larger for the same reason that marginal cost is more elastic to output there.

This redistribution is not neutral across household types. Constrained households receive labour income alone, and it is labour income that bears the whole of the lower-nest adjustment; savers receive the profits that the same price rigidity creates. A contractionary monetary policy shock therefore transfers resources from wages to profits, and in this model the size of that transfer is increasing in the robot income share. The channel is familiar from [Bilbiie \(2008\)](#), where counter-cyclical profits accruing to unconstrained households are what amplify the consumption response of the constrained. What automation adds is that the transfer grows with the robot cost share, so the amplification is not a fixed feature of the economy but a function of how automated it is.

The incidence result of Section 4.5 therefore has two sources rather than one. The first is the wage channel of Section 4.4, which determines how far unskilled wages fall. The second is the profit channel described here, which determines how much of what wages lose is received by households on the other side of the participation margin. Both scale with the robot cost share, and both do so because the predeterminacy of the robot stock makes marginal cost more responsive to output. That the two channels share a single origin is why the distributional and aggregate results of this section cannot be separated: the same rigidity that steepens the inflation–output trade-off is what widens the consumption differential.

4.7 Dynamics

The impact responses discussed above are also the peak responses for every variable except robot investment, as Figure 4 shows. Output stands at forty percent of its impact value one quarter later and at seven percent after three quarters, and the paths are monotone thereafter. The interest rate rule contains a smoothing coefficient of 0.8, but the effective decay of the responses is closer to 0.4 per quarter, because the systematic component of the rule pulls against the smoothing while inflation is below target. The dynamics should therefore be described as a fast monotone decay rather than as endogenous propagation, and the impact table carries the substance of this section.

4.8 Sensitivity to the normalisation

The aggregate result is robust to the specification changes considered. The impact output cost of disinflation falls by thirty-seven percent over the range of robot income shares from five to twenty-

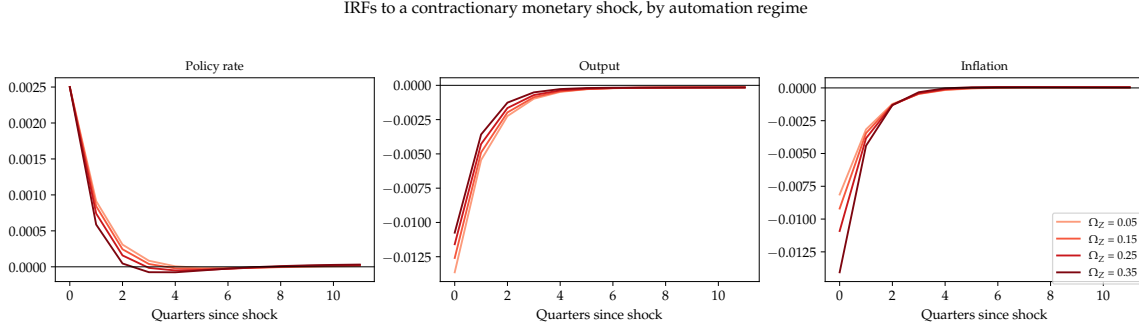


Figure 4: IRFs to a contractionary monetary shock

five percent, and by a comparable proportion without investment adjustment costs, across the range of the lower-nest elasticity, and in the counterfactual economy with no constrained households at all. Removing interest rate smoothing steepens the decline to forty-one percent, so the baseline specification understates rather than manufactures the gradient. That the result survives changes on four independent margins, including the removal of the feature that the two-agent structure exists to represent, is a stronger statement about the mechanism than any single specification could provide.

The distributional results are sensitive to how regimes are compared, and the sensitivity is instructive. Normalising each regime by a common structural innovation rather than by a common impact policy-rate response makes Ricardian consumption appear to fall progressively less as automation deepens, which would invite the reading that savers become insulated. Under the normalisation adopted here, in which each regime is scaled to deliver the same twenty-five basis point impact response of the policy rate, Ricardian consumption is flat across regimes to the third decimal place. The widening of the consumption differential is therefore entirely a hand-to-mouth phenomenon driven by the wage channel of Section 4.4, and not a movement at both ends of the distribution. Delivering the same policy action requires a twenty-eight percent larger innovation at the highest regime than at the lowest, because inflation responds more strongly there and the rule's own feedback partly undoes the shock, which is why the two normalisations diverge.

5 Policy implications

The results reported above are positive statements about transmission, and what they license by way of prescription has to be drawn carefully. This section separates two questions. The first is what the model is entitled to say at all, given that it measures the incidence of a policy action rather than the welfare consequences of automation itself. The second is what the interaction between the automation margin and the participation margin implies for the division of labour between the monetary instrument and the instruments that reach household balance sheets directly.

5.1 What the model measures

The results of Section 4 admit a reading that is more useful than either of its halves taken alone. The aggregate finding is that disinflation becomes cheaper in output terms as automation deepens; the distributional finding is that the burden of that disinflation shifts towards households who cannot smooth consumption. These are not two results about an automating economy. They are one result, because both follow from the same feature of the production structure: the robot stock is predetermined, so a larger robot cost share means a larger share of the short-run adjustment falls on the one input that can still move, and that input is supplied disproportionately by constrained households. Disinflation becomes cheaper in the aggregate precisely to the extent that it is borne by those least able to bear it.

This suggests a way of using the model that is not available from either result on its own. Automation intensity is observable, at least as an indicator, and the model maps it onto two quantities of direct interest to a policymaker: the output cost of a given disinflation, and the excess consumption burden borne by households excluded from asset markets. Neither quantity is unfamiliar. What the model supplies is the link between them and a third variable that distributional analyses of monetary policy do not ordinarily carry, namely the automation intensity of production.

That omission matters because it makes the distributional footprint of monetary policy look more stable than it is. Analyses of the incidence of monetary policy generally take the participation margin as a structural feature of the economy, so the burden borne by constrained households is read off a household structure that is assumed to hold fixed over the horizon of interest. The results of Section 4.5 imply that this is not sufficient. The excess burden roughly doubles across the range of robot income shares considered here at an unchanged share of constrained households, because the exposure operates through the production structure rather than through the household structure. An economy can therefore become materially more unequal in its exposure to monetary policy without any change in who participates in asset markets.

The point is sharper still when set against the aggregate result, since the two move in opposite directions and a conventional objective function sees only one of them. A central bank assessing automation through the sacrifice ratio concludes that its task is becoming easier: the same disinflation can be achieved at declining cost to output. That conclusion is correct on its own terms and incomplete, since a quadratic loss defined over the output gap and inflation aggregates across households by construction and is therefore blind to the reallocation that makes the improvement possible. An automating economy does not become easier to stabilise; it becomes an economy in which the cost of stabilisation is less visible in the aggregate statistics used to evaluate it.

5.2 Two margins, two instruments

The implication would be uncomfortable if the two margins were the same margin, since the authority would then face a choice between aggregate and distributional objectives with no instrument to separate them. They are not, though the sense in which they are distinct requires care. Varying the share of constrained households leaves the aggregate result untouched: the decline in the sacrifice ratio across automation regimes is the same at every participation rate considered, including the limiting case of none. It also leaves the structure of the distributional result untouched, in that the excess burden expressed relative to the saver loss, and its gradient in the robot income share, are close to invariant. Automation intensity governs the gradient of the distributional cost; the participation share governs its level.

The two do not, however, enter that level independently. The excess burden rises more than proportionally with the share of constrained households, and the factor by which it rises is itself increasing in the robot income share, so the two margins are complements rather than separable multiplicative terms. The reading for policy is more constructive than the multiplicative case would have been, not less. Measures that widen participation in asset markets reduce the excess burden documented in Section 4.5 without altering the aggregate trade-off, and they reduce it by more, not less, the more automated the economy in which they are introduced. Conversely, the improvement in the sacrifice ratio is available whatever the participation rate. The two policy problems raised by automation are addressable independently even though the quantities they act on interact.

What monetary policy itself should take from this is narrower but not trivial. The remedy for the distributional cost lies outside the monetary authority's instrument set, but the cost itself belongs in its information set. The results imply that the incidence of a given tightening is a function of the economy's automation intensity, so the same policy action carries a different distributional footprint at different points in an automating economy's development. An authority that regards that footprint as a constraint rather than as an objective still needs to know that it is growing, and the aggregate statistics on which the assessment would ordinarily rest are precisely the ones that conceal it.

6 Conclusions

This paper has asked how the degree of automation alters the transmission of monetary policy to output and inflation, and how it alters the incidence of that transmission across household types. The two questions have one answer. Robotic capital is predetermined with respect to the shock, so the larger the share of production it accounts for, the larger the fraction of the short-run adjustment in marginal cost that must be borne by the margins still free to move within the period. On the price side this is the rental rate on the fixed stock, whose weight in marginal cost rises with the robot share; on the quantity side it is unskilled labour. For the aggregate, this means inflation

carries more of the response and output less, so the output cost of a given disinflation falls as automation deepens. For the distribution, it means that the households supplying that labour, and in part excluded from the asset markets that would allow them to smooth, absorb a progressively larger share of the cost. A counterfactual economy in which every household participates in asset markets delivers the same steepening and the same divergence between skilled and unskilled wages, so the aggregate half of the result owes nothing to limited participation: the asset market constraint governs the level of the distributional cost, not its gradient.

Two implications for the existing literature follow. The first concerns the sign of the aggregate result. Frameworks in which the automated margin adjusts contemporaneously with the policy shock find that automation flattens the relationship between inflation and real activity; with the automated input predetermined, the relationship steepens instead. Stated in that form the difference is a discriminating assumption rather than a contradiction, and one on which evidence can be brought to bear. Where the automated stock is predetermined but the reported object is an average Phillips-curve slope rather than an impact response, there is no disagreement at all, only a difference in what is measured. The second concerns how incidence is read. A model of this kind is usually taken to deliver distributional results because it restricts asset market participation, and the aggregate result to be a consequence of that restriction. Here the order runs the other way. The steepening is a production-side property that survives intact when every household participates in asset markets, and the constraint governs how much of the resulting burden a given household absorbs rather than whether the burden exists. Limited participation is therefore the wrong place to look for the aggregate consequences of automation, and the right place to look for its distributional ones.

The identification of the model bounds these claims unevenly, and the boundary is worth stating precisely. Indexing regimes by the robot income share rather than by the share parameter of the production function pins the cost share of robots in the lower nest, and with it the amplification factor through which the mechanism operates. That factor is identical at every value of the elasticity of substitution between robots and unskilled labour, so the aggregate trade-off and the mechanism that generates it are immune by construction to a parameter the steady state is unable to discipline. What the calibration target does not pin is the division of the constrained burden between skilled and unskilled households. The elasticity enters the dynamics through the conditional factor demands without leaving any trace in the steady-state shares the calibration targets, and over the range we consider the skill gap moves by an order of magnitude while the burden of constrained households relative to savers moves hardly at all. The direction and the ranking of every claim made here hold throughout; the magnitude of the one claim specific to this paper waits on a measured elasticity, and the model's contribution on this point is to identify which quantity would have to be measured. Relative risk aversion plays an analogous role on the household side, since the sign of the constrained hours response turns on whether it lies below unity, and the value at which it does not is the value at which the income effect reverses a response the empirical literature reports with the opposite sign.

Three restrictions of scope qualify the magnitudes reported. The first is that the grid is not a set of observed economies: the robot income shares considered are not intended to match those implied by current robot stocks, so the exercise identifies the direction in which automation moves the

transmission of monetary policy rather than the position of any economy along that path, and it is the gradient across regimes rather than the level at any one of them that carries the result. The steady-state skill premium is held fixed across regimes, which is a *ceteris paribus* restriction rather than a prediction, since the literature on automation generally finds that robot adoption raises skill inequality. What the paper's scope adds here is that the direction of the resulting error is known: because the automation of routine cognitive tasks lies outside the model, the only automating force acting on the premium is one that substitutes for unskilled labour, so the movement suppressed by the restriction can only go one way and the distributional magnitudes reported are a lower bound. Were cognitive automation in scope, skilled labour would face a substitutable margin of its own and the sign of that omission would be ambiguous. Finally, consumption is pooled across skill types within the saver household, so the model speaks to three consumption responses rather than four.

A fourth restriction concerns the margin closest to the mechanism. Capital services are equal to the installed stock in this model, with no variable rate of utilisation, and this is the assumption on which (29) turns rather than a modelling convenience. Utilisation is precisely the channel through which a predetermined stock becomes partly variable within the period, so a specification in which robot services could be varied by running the installed stock harder would weaken the amplification factor and, in the limit, remove it. The restriction is deliberate and reflects the object the paper studies. Industrial robots in the sense of the ISO definition are configured for particular tasks and lines, and the short-run margin available on an installed robot stock is closer to fixed than the utilisation margin conventionally assigned to aggregate capital; a model in which automation capital can be run harder within the quarter is a model of a different input, and the quantitative literature that adopts such a margin reaches the opposite sign for that reason. The claim made here is therefore conditional on predeterminacy rather than a general claim about automation, and it is stated as such.

The reading for policy is positive throughout. Nothing established here licenses a different response to a given shock, and the remedy for the distributional cost documented above lies outside the monetary authority's instrument set in any case. What the results imply is narrower and concerns the information on which an assessment of automation is made. The incidence of an unchanged policy action is a function of the automation intensity of production, a variable that analyses of the distributional footprint of monetary policy do not ordinarily carry, and an economy can therefore become materially more unequal in its exposure to policy with no change in who participates in asset markets. An authority that judges the consequences of automation by the sacrifice ratio observes an improvement, and the improvement is real; what the aggregate statistic cannot show is that it is produced by the reallocation of the burden of stabilisation towards the households whose consumption the instrument reaches last.

Three extensions follow directly. The first is to replace the binary participation margin with a distribution of marginal propensities to consume in a heterogeneous-agent setting. The aggregate result should carry over, since it survives the removal of constrained households entirely, but the level of the distributional cost is exactly what a binary margin represents coarsely, and the complementarity between the participation and automation margins documented in Section 5.2 suggests that a richer distribution would sharpen rather than dilute it. The second is estimation. The elas-

ticity identified above as the parameter carrying the distributional magnitude is the natural object, and the same exercise offers a test of the assumption on which the aggregate result rests, since the responsiveness of the installed robot stock over the horizon of monetary transmission is the quantity that separates our conclusion from the flattening reported elsewhere. The third, and the one that would test the mechanism most directly, is the extensive margin. Employment adjusts here only through hours supplied at a wage that clears the market, so the whole incidence of a contraction on constrained households arrives as a smaller wage bill on work that continues rather than as separation from work. That is the narrower of the two channels through which automation is generally argued to reach the unskilled, and it is the one to which a household unable to smooth is least exposed. Introducing search and matching would place part of the adjustment on who works rather than on how much, and it would also change the route by which automation reaches the wage: in [Leduc and Liu \(2024\)](#) and [Basso and Rachedi \(2025\)](#) the substitutability of machines for workers operates through the outside option in bargaining rather than through the conditional factor demands of a technology, so the two routes are alternatives rather than components of a single channel. Whether the incidence documented here survives that substitution, and whether it is sharpened once constrained households face a risk of separation alongside a fall in the wage, is the natural test of how far the result depends on the intensive margin on which it is built.

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A Model equations and derivations

This appendix collects the derivations omitted from Section 3 and states the complete non-linear system. Nothing here is required to follow the argument of the main text; it is provided so that every equation entering the numerical solution can be traced to a primitive.

A.1 Ricardian households

The representative Ricardian family of mass $1-\lambda$ chooses $\{C_t^r, N_{S,t}^r, N_{U,t}^r, I_{K,t}, I_{Z,t}, K_{t+1}, Z_{t+1}, B_{t+1}\}$ to maximise, per member,

$$E_t \sum_{i=0}^{\infty} \beta^i \left[\frac{(C_{t+i}^r)^{1-\gamma}}{1-\gamma} - n_S \omega_S \frac{(N_{S,t+i}^r)^{1+\varphi}}{1+\varphi} - n_U \omega_U \frac{(N_{U,t+i}^r)^{1+\varphi}}{1+\varphi} \right]$$

subject to the budget constraint, the two accumulation equations and the adjustment cost function (1) of the main text. Let $\Lambda_t = (C_t^r)^{-\gamma}$ and let $\Lambda_t Q_{K,t}$ and $\Lambda_t Q_{Z,t}$ be the multipliers on the accumulation equations, so that $Q_{j,t}$ is the shadow value of a unit of installed capital of type j in consumption units. Write $x_{j,t} \equiv I_{j,t}/I_{j,t-1}$.

The conditions with respect to C_t^r , B_{t+1} and the two labour supplies are standard and give (3), (8) and (9). The condition with respect to $I_{K,t}$ requires the derivative of installed capital with respect to the current investment flow, which enters at two dates. At t the term $[1 - S_K(x_{K,t})] I_{K,t}$ contributes $1 - S_K(x_{K,t}) - S'_K(x_{K,t})x_{K,t}$, since $\partial x_{K,t}/\partial I_{K,t} = 1/I_{K,t-1}$. At $t+1$ the flow $I_{K,t}$ enters the denominator of $x_{K,t+1}$, contributing $S'_K(x_{K,t+1})x_{K,t+1}^2$ once the negative sign of $\partial x_{K,t+1}/\partial I_{K,t}$ is carried through. Setting the sum to zero against the unit purchase price gives (4), and the same construction with the purchase price p_t^Z gives (5). The conditions with respect to K_{t+1} and Z_{t+1} give (6) and (7) directly.

The within-family weights n_S and n_U cancel from the two labour supply conditions because each weights the disutility of a member's hours and that member's contribution to family income identically. This is what permits a single consumption variable to serve both skill types while hours differ.

A.2 Log-linearisation of the Ricardian block

Because $S_j(1) = S'_j(1) = 0$ and $S''_j(1) = \tau_j$, the adjustment cost and its first derivative vanish at the deterministic steady state and only the second derivative survives a first-order expansion. Writing $\hat{x}_{j,t} = \hat{I}_{j,t} - \hat{I}_{j,t-1}$ and using $S'_j(x_{j,t}) \simeq \tau_j \hat{x}_{j,t}$, condition (4) becomes $0 = \hat{Q}_{K,t} - \tau_K \hat{x}_{K,t} + \beta \tau_K \mathbb{E}_t \hat{x}_{K,t+1}$, and (5) the same with $\hat{Q}_{Z,t} - \hat{p}_t^Z$ in place of $\hat{Q}_{K,t}$, since $Q_Z^{ss} = \bar{p}^Z$. Rearranging gives the two investment conditions listed in Table 6.

The valuation conditions are linearised using $\beta(\bar{r}^K + 1 - \delta_k) = 1$, which implies $\beta \bar{r}^K = 1 - \beta(1 - \delta_k)$ and so weights the rental and the resale value by $1 - \beta(1 - \delta_k)$ and $\beta(1 - \delta_k)$ respectively. Together with $\hat{\Lambda}_t = -\gamma \hat{C}_t^r$ this gives the two conditions in Table 6, each of which takes the form of an Euler equation in which the return on the stock replaces the bond rate and includes a capital gain in its own shadow price. Setting $\tau_K = \tau_Z = 0$ delivers $\hat{Q}_{K,t} = 0$ and $\hat{Q}_{Z,t} = \hat{p}_t^Z$ and collapses both to rental-only Euler equations.

The accumulation equations linearise to the identities they would take without adjustment costs, since the cost term is second order, so the friction enters the dynamics through the investment conditions alone and leaves the capital stocks' laws of motion untouched. The lagged flows $I_{K,t-1}$ and $I_{Z,t-1}$ therefore enter the state vector as carriers rather than as behavioural objects, and are implemented as the two identities $\hat{I}_{j,t}^{lag} = \hat{I}_{j,t-1}$.

A.3 Hand-to-mouth households

Substituting the budget constraint $C_{j,t}^h = (W_{j,t}/P_t)N_{j,t}^h$ into the intratemporal condition (11) eliminates consumption and gives $\omega_j(N_{j,t}^h)^{\varphi+\gamma} = (W_{j,t}/P_t)^{1-\gamma}$, from which the closed forms (12) follow by taking the $(\varphi + \gamma)$ -th root and substituting back. The elasticities $\frac{1-\gamma}{\varphi+\gamma}$ and $\frac{1+\varphi}{\varphi+\gamma}$ are exact rather than approximations, since the closed forms are already log-linear in the real wage. In the numerical system the budget constraint and the intratemporal condition are carried separately rather than substituted, which is equivalent and leaves hours and consumption as distinct variables for reporting.

A.4 Final-good firms

The final-good firm chooses $\{Y_t(i)\}$ to minimise $\int_0^1 P_t(i)Y_t(i) di$ subject to the CES aggregator. The first-order condition equates the ratio of prices to the ratio of marginal products, giving $Y_t(i) = (P_t(i)/P_t)^{-\varepsilon_p} Y_t$ with P_t defined as the multiplier on the constraint. Substituting the demand back into the aggregator confirms that this multiplier is the price index $P_t = \left[\int_0^1 P_t(i)^{1-\varepsilon_p} di \right]^{1/(1-\varepsilon_p)}$,

and substituting it into the expenditure identity confirms zero profits for the final-good sector.

A.5 Constant returns and the common marginal cost

Both nests are homogeneous of degree one in their arguments, the outer layer is Cobb–Douglas and A_t is Hicks-neutral, so the technology $Y_t(i) = A_t G(\bar{K}_t, \bar{N}_{S,t}, \bar{Z}_t, \bar{N}_{U,t})$ is homogeneous of degree one as a whole: scaling every input by κ scales output and the cost bill by κ . The solution to the cost-minimisation problem therefore has the input mix depending on relative factor prices alone and the input levels proportional to output, so total variable cost is linear in output and takes the form $MC_t Y_t(i)$, with marginal and average variable cost coinciding and neither depending on the scale at which the firm operates. Because factor markets are competitive, every intermediate producer faces the same factor prices and the same A_t , so the unit cost MC_t is common across i . This is what permits the pricing problem to be solved once for a representative resetting firm, and it is what makes every resetting firm choose the same price. The fixed cost Φ is a constant in final-good units and shifts total but not marginal cost, leaving this argument intact.

A.6 Cost minimisation, nest by nest

In the lower nest the firm minimises $r_t^Z \bar{Z}_t + w_{U,t} \bar{N}_{U,t}$ subject to $L_t = [\theta_Z \bar{Z}_t^{\rho_L} + (1 - \theta_Z) \bar{N}_{U,t}^{\rho_L}]^{1/\rho_L}$. The ratio of the two first-order conditions gives the expansion path $\bar{Z}_t / \bar{N}_{U,t} = \left[\frac{\theta_Z}{1 - \theta_Z} \frac{w_{U,t}}{r_t^Z} \right]^{\sigma_L}$, and substituting it into the aggregator yields the unit cost (15) of the main text together with the conditional demands

$$\bar{Z}_t = L_t \theta_Z^{\sigma_L} \left(\frac{r_t^Z}{P_{L,t}} \right)^{-\sigma_L}, \quad \bar{N}_{U,t} = L_t (1 - \theta_Z)^{\sigma_L} \left(\frac{w_{U,t}}{P_{L,t}} \right)^{-\sigma_L}. \quad (30)$$

The same construction in the upper nest, with L_t entering at price $P_{L,t}$ alongside skilled labour at $w_{S,t}$, gives (16) and

$$\bar{N}_{S,t} = X_t \theta_S^{\sigma_X} \left(\frac{w_{S,t}}{P_{X,t}} \right)^{-\sigma_X}, \quad L_t = X_t (1 - \theta_S)^{\sigma_X} \left(\frac{P_{L,t}}{P_{X,t}} \right)^{-\sigma_X}. \quad (31)$$

The outer Cobb–Douglas layer combines $\bar{K}_t$ at r_t^K with X_t at $P_{X,t}$, delivering the factor demands $r_t^K \bar{K}_t = \alpha MC_t Y_t$ and $P_{X,t} X_t = (1 - \alpha) MC_t Y_t$ and the unit cost (17).

Log-linearising the three unit cost functions and applying Shephard's lemma gives $\hat{P}_{L,t} = s_Z \hat{r}_t^Z + (1 - s_Z) \hat{w}_{U,t}$, $\hat{P}_{X,t} = s_S \hat{w}_{S,t} + (1 - s_S) \hat{P}_{L,t}$ and $\hat{m}\hat{c}_t = -\hat{A}_t + \alpha \hat{r}_t^K + (1 - \alpha) \hat{P}_{X,t}$, which compose to (21). Each factor price enters marginal cost weighted by the product of the cost shares along the path from its nest to the top of the technology. The conditional demands linearise to the

constant-elasticity forms listed in Table 6, in which each factor is demanded in proportion to its nest aggregate and falls in its own price relative to the unit cost of that nest with elasticity σ_L or σ_X .

Because the population shares are constants, $\hat{\bar{K}}_t = \hat{K}_t$ and $\hat{\bar{Z}}_t = \hat{Z}_t$, so the aggregate and per-member capital stocks are the same variable in log deviations and the system carries only one of each. The production function itself is not imposed as a separate log-linear equation: given constant returns, the two outer factor demands and the definitions of the unit costs imply $\hat{Y}_t = \hat{A}_t + \alpha \hat{K}_t + (1 - \alpha) \hat{X}_t$, which is used as a numerical check on the solution rather than as an equation of the system.

A.7 Pricing

A firm resetting in period t chooses $P_t(i)$ to maximise the expected discounted value of future operating profits, the fixed cost being independent of the price and therefore absent from the first-order condition:

$$\mathbb{E}_t \left[\sum_{s=0}^{\infty} (\beta \theta_p)^s \xi_{t,t+s} \left(\frac{\Gamma_{t,t+s} P_t(i)}{P_{t+s}} - MC_{t+s} \right) \left(\frac{\Gamma_{t,t+s} P_t(i)}{P_{t+s}} \right)^{-\varepsilon_p} Y_{t+s} \right],$$

where $\Gamma_{t,t+s} \equiv \Pi_{t-1,t+s-1}^{\zeta_p}$ is the indexation factor and $\xi_{t,t+s} \equiv (C_{t+s}^r / C_t^r)^{-\gamma}$ the ratio of Ricardian marginal utilities, firms being owned by Ricardian households alone. Differentiating with respect to $P_t(i)$ and using the fact that real marginal cost is common across firms gives the reset condition

$$\mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_p)^s \xi_{t,t+s} Y_{t+s} \left(\frac{P_t^*}{P_{t+s}} \Gamma_{t,t+s} \right)^{-\varepsilon_p} \left(\frac{P_t^*}{P_{t+s}} \Gamma_{t,t+s} - \frac{\varepsilon_p}{\varepsilon_p - 1} MC_{t+s} \right) = 0, \quad (32)$$

in which the reset price is a discounted weighted average of current and expected future marginal costs marked up by $\varepsilon_p / (\varepsilon_p - 1)$. Under flexible prices only the $s = 0$ term survives and the condition collapses to the constant-markup benchmark $P_t^* / P_t = \frac{\varepsilon_p}{\varepsilon_p - 1} MC_t$ that underlies (19). The aggregate price index, combining resetting and indexing firms, obeys

$$1 = (1 - \theta_p) \left(\frac{P_t^*}{P_t} \right)^{1-\varepsilon_p} + \theta_p \left(\frac{\Pi_{t-1}^{\zeta_p}}{\Pi_t} \right)^{1-\varepsilon_p}, \quad \Pi_t \equiv \frac{P_t}{P_{t-1}}. \quad (33)$$

A.8 The fixed cost and the resource constraint

Aggregating (18) over the symmetric continuum, with price dispersion equal to unity to first order, gives $D_t = (1 - MC_t)Y_t - \Phi$. In the symmetric flexible-price steady state the reset-price condition sets $MC^{ss} = (\varepsilon_p - 1)/\varepsilon_p$, and imposing $D^{ss} = 0$ pins Φ to the steady-state operating surplus, as in (19). Zero steady-state profit does not imply zero fluctuation: differentiating $D_t = (1 - MC_t)Y_t - \Phi$ at $D^{ss} = 0$ gives $dD_t/Y^{ss} = (1 - MC^{ss})\hat{Y}_t - MC^{ss}\hat{m}\hat{c}_t$, so dividend income varies at first order and accrues entirely to Ricardian households.

The fixed cost must appear in the goods market clearing condition (24) for that condition to be consistent with (19). Under constant returns, factor payments exhaust $MC_t Y_t$, so household income is $MC_t Y_t + D_t$; with $D^{ss} = 0$ that income finances $C^{ss} + (1 - \lambda)(I_K^{ss} + \bar{p}^Z I_Z^{ss})$ exactly, leaving $\Phi = (1 - MC^{ss})Y^{ss}$ as the residual real resource use. Omitting Φ from (24) would leave the constraint unbalanced by exactly that amount. Since Φ is a constant in final-good units it has no deviation from steady state and contributes no term to the log-linear resource constraint, which is why the three expenditure shares sum to $1 - 1/\varepsilon_p$ rather than to unity.

A.9 The complete non-linear system

Table 5 collects the equilibrium conditions. The endogenous variables are $C_t^r, C_{S,t}^h, C_{U,t}^h, C_t, N_{S,t}^r, N_{U,t}^r, N_{S,t}^h, N_{U,t}^h, \bar{N}_{S,t}, \bar{N}_{U,t}, K_t, Z_t, I_{K,t}, I_{Z,t}, Q_{K,t}, Q_{Z,t}, r_t^K, r_t^Z, w_{S,t}, w_{U,t}, L_t, X_t, P_{L,t}, P_{X,t}, Y_t, MC_t, D_t, \Pi_t, P_t^*$ and R_t , together with the exogenous processes A_t and p_t^Z .

Condition	Description
$(C_t^r)^{-\gamma} = \beta R_t \mathbb{E}_t \left[(C_{t+1}^r)^{-\gamma} \frac{P_t}{P_{t+1}} \right]$	Bond Euler equation
$\omega_j (N_{j,t}^r)^\varphi (C_t^r)^\gamma = w_{j,t}, j \in \{S, U\}$	Ricardian labour supply
$1 = Q_{K,t} [1 - S_K - S'_K x_{K,t}] + \beta \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} Q_{K,t+1} S'_K x_{K,t+1}^2 \right]$	Investment condition, traditional capital
$p_t^Z = Q_{Z,t} [1 - S_Z - S'_Z x_{Z,t}] + \beta \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} Q_{Z,t+1} S'_Z x_{Z,t+1}^2 \right]$	Investment condition, robotic capital
$Q_{j,t} = \beta \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} (r_{t+1}^j + (1 - \delta_j) Q_{j,t+1}) \right], j \in \{K, Z\}$	Valuation of installed capital
$K_{t+1} = (1 - \delta_K) K_t + [1 - S_K(x_{K,t})] I_{K,t}$	Accumulation, traditional capital
$Z_{t+1} = (1 - \delta_Z) Z_t + [1 - S_Z(x_{Z,t})] I_{Z,t}$	Accumulation, robotic capital
$C_{j,t}^h = w_{j,t} N_{j,t}^h, j \in \{S, U\}$	Hand-to-mouth budget constraint
$\omega_j (N_{j,t}^h)^\varphi (C_{j,t}^h)^\gamma = w_{j,t}, j \in \{S, U\}$	Hand-to-mouth labour supply

Condition	Description
$\bar{N}_{S,t} = (1 - \mu) [\lambda_S N_{S,t}^h + (1 - \lambda_S) N_{S,t}^r]$	Aggregate skilled labour
$\bar{N}_{U,t} = \mu [\lambda_U N_{U,t}^h + (1 - \lambda_U) N_{U,t}^r]$	Aggregate unskilled labour
$\bar{K}_t = (1 - \lambda) K_t, \quad \bar{Z}_t = (1 - \lambda) Z_t$	Aggregate capital stocks
$L_t = [\theta_Z \bar{Z}_t^{\rho_L} + (1 - \theta_Z) \bar{N}_{U,t}^{\rho_L}]^{1/\rho_L}$	Lower nest
$X_t = [\theta_S \bar{N}_{S,t}^{\rho_X} + (1 - \theta_S) L_t^{\rho_X}]^{1/\rho_X}$	Upper nest
$Y_t = A_t \bar{K}_t^\alpha X_t^{1-\alpha}$	Production function
$\bar{Z}_t = L_t \theta_Z^{\sigma_L} (r_t^Z / P_{L,t})^{-\sigma_L}, \quad \bar{N}_{U,t} = L_t (1 - \theta_Z)^{\sigma_L} (w_{U,t} / P_{L,t})^{-\sigma_L}$	Lower-nest factor demands
$\bar{N}_{S,t} = X_t \theta_S^{\sigma_X} (w_{S,t} / P_{X,t})^{-\sigma_X}, \quad L_t = X_t (1 - \theta_S)^{\sigma_X} (P_{L,t} / P_{X,t})^{-\sigma_X}$	Upper-nest factor demands
$r_t^K \bar{K}_t = \alpha MC_t Y_t, \quad P_{X,t} X_t = (1 - \alpha) MC_t Y_t$	Outer factor demands
$MC_t = A_t^{-1} (r_t^K / \alpha)^\alpha (P_{X,t} / (1 - \alpha))^{1-\alpha}$	Real marginal cost
Equation (32)	Optimal reset price
$1 = (1 - \theta_p) (P_t^* / P_t)^{1-\varepsilon_p} + \theta_p (\Pi_{t-1}^{\zeta_p} / \Pi_t)^{1-\varepsilon_p}$	Price index
$D_t = (1 - MC_t) Y_t - \Phi, \quad \Phi = Y^{ss} / \varepsilon_p$	Dividends and fixed cost
$C_t = (1 - \lambda) C_t^r + (1 - \mu) \lambda_S C_{S,t}^h + \mu \lambda_U C_{U,t}^h$	Aggregate consumption
$Y_t = C_t + (1 - \lambda) (I_{K,t} + p_t^Z I_{Z,t}) + \Phi$	Goods market clearing
$B_t = 0$	Bond market clearing
Equation (27)	Monetary policy rule
$\log p_t^Z = (1 - \rho_p) \log \bar{p}^Z + \rho_p \log p_{t-1}^Z + \varepsilon_{p,t}$	Relative price of robotic investment
$\log A_t = \rho_A \log A_{t-1} + \varepsilon_t^A$	Total factor productivity

Table 5: The non-linear equilibrium system.

B Computational specification

The model is solved entirely in Python. The steady state is solved with `scipy.optimize.root` and the log-linear system with `scipy.linalg.ordqz`, which supplies the ordered generalised Schur decomposition required by the method of Klein (2000); `numpy` is used for the linear algebra and `pandas` and `matplotlib` for the reporting of impulse responses. No proprietary software

or dedicated model-solving package is used, so the system is written out equation by equation in the coefficient matrices of Section B.3 rather than parsed from a model file. The code is organised as three modules: one solving the steady state and holding the calibration, one building the coefficient matrices and computing impulse responses, and one running the regime experiments and the sensitivity exercises reported in Appendix A onwards. Version numbers for the interpreter and the libraries are recorded in the replication package.

B.1 Steady state

The deterministic steady state is solved as a system of eight equations in the eight unknowns $(Z, K, N_S^r, N_U^r, \theta_Z, \theta_S, w_S, w_U)$. Four conditions are behavioural and four are calibration targets or consistency requirements. The two rentals are known in closed form from the household problem, $\bar{r}^K = 1/\beta - (1 - \delta_k)$ and $\bar{r}^Z = \bar{p}^Z [1/\beta - (1 - \delta_z)]$, since the adjustment costs are inoperative at $x_K = x_Z = 1$; the residuals therefore require the rentals implied by the technology to equal these values. The two Ricardian labour supply conditions are imposed at the trial wages and the implied Ricardian consumption. The skill premium target $\Upsilon = w_S/w_U$ and the robot income share target $\Omega_Z = \bar{r}^Z \bar{Z}/Y$ pin the two share parameters θ_S and θ_Z . The last two residuals require the wages implied by the upper- and lower-nest conditional demands to equal the trial wages, which closes the loop running from wages to hand-to-mouth hours to the labour aggregates and back to factor prices.

Hand-to-mouth hours and consumption are not unknowns of the system: at any trial wage they follow in closed form from (12), which is what keeps the system at eight equations despite the four household cells. Ricardian consumption is likewise computed rather than solved for, as labour income plus the net return on the two stocks, using $D^{ss} = 0$.

Four normalisations are imposed. Total factor productivity is set to unity, the mean relative price of robotic investment is set to unity so that s_Z carries the whole of its influence, steady-state dividends are set to zero through the fixed cost, and a single labour disutility scale ω is used for all four household cells, so that no part of the steady-state dispersion between cells comes from preference heterogeneity. The benchmark further imposes $\lambda_S = \lambda_U = \lambda$, under which the within-Ricardian population weights collapse to $n_S = 1 - \mu$ and $n_U = \mu$.

The system is solved with the modified Powell hybrid method of `scipy.optimize.root` to a residual tolerance of 10^{-8} , with the Jacobian evaluated by forward differences rather than supplied analytically.

B.2 Coefficients entering the log-linear system

Beyond the parameters, the log-linear system is a function of a set of steady-state shares computed from the solution above. The consumption weights χ_r , χ_{hS} and χ_{hU} and the expenditure shares s_C , s_{I_K} and s_{I_Z} are defined in Section 3.3, alongside (25) and (26), and the two nest-level cost shares s_S and s_Z in Section 3.2.2, alongside (21). The labour aggregation weights are

$$\nu_{jh} \equiv \frac{\lambda_j N_j^h}{\lambda_j N_j^h + (1 - \lambda_j) N_j^r}, \quad \nu_{jr} \equiv 1 - \nu_{jh}, \quad j \in \{S, U\},$$

in which the skill population shares μ and $1 - \mu$ cancel, so that the composition of each skill group's hours depends on the participation shares and on relative hours within the group alone. The Phillips curve coefficients are $\kappa_f \equiv \beta / (1 + \beta \zeta_p)$ and $\kappa_p \equiv (1 - \theta_p)(1 - \beta \theta_p) / [\theta_p(1 + \beta \zeta_p)]$, which reduce to β and $(1 - \theta_p)(1 - \beta \theta_p) / \theta_p$ at $\zeta_p = 0$.

B.3 The complete log-linear system

Table 6 lists the thirty-three equations of the log-linear system in the order in which they are coded. Seven variables are predetermined, namely $\hat{A}_t$, $\hat{p}_t^Z$, $\hat{K}_t$, $\hat{Z}_t$, $\hat{I}_{K,t}^{lag}$, $\hat{I}_{Z,t}^{lag}$ and $\hat{R}_t^{lag}$, and twenty-six are jump variables. The last three equations of the table are identities carrying lagged values into the state vector rather than behavioural relations. Aggregate and per-member capital stocks coincide in log deviations, so $\hat{\hat{K}}_t$ and $\hat{\hat{Z}}_t$ are written $\hat{K}_t$ and $\hat{Z}_t$ throughout.

Equation	Description
$\hat{C}_t^r = \mathbb{E}_t \hat{C}_{t+1}^r - \frac{1}{\gamma} \left(\hat{R}_t - \mathbb{E}_t \hat{\pi}_{t+1} \right)$	Bond Euler equation
$\varphi \hat{N}_{S,t}^r + \gamma \hat{C}_t^r = \hat{w}_{S,t}$	Ricardian labour supply, skilled
$\varphi \hat{N}_{U,t}^r + \gamma \hat{C}_t^r = \hat{w}_{U,t}$	Ricardian labour supply, unskilled
$\hat{C}_{S,t}^h = \hat{w}_{S,t} + \hat{N}_{S,t}^h$	Hand-to-mouth budget, skilled
$\varphi \hat{N}_{S,t}^h + \gamma \hat{C}_{S,t}^h = \hat{w}_{S,t}$	Hand-to-mouth labour supply, skilled
$\hat{C}_{U,t}^h = \hat{w}_{U,t} + \hat{N}_{U,t}^h$	Hand-to-mouth budget, unskilled
$\varphi \hat{N}_{U,t}^h + \gamma \hat{C}_{U,t}^h = \hat{w}_{U,t}$	Hand-to-mouth labour supply, unskilled
$\hat{Q}_{K,t} = \tau_K \left(\hat{I}_{K,t} - \hat{I}_{K,t-1} \right) - \beta \tau_K \left(\mathbb{E}_t \hat{I}_{K,t+1} - \hat{I}_{K,t} \right)$	Investment condition, traditional capital

Equation	Description
$\hat{Q}_{Z,t} = \hat{p}_t^Z + \tau_Z (\hat{I}_{Z,t} - \hat{I}_{Z,t-1}) - \beta \tau_Z (\mathbb{E}_t \hat{I}_{Z,t+1} - \hat{I}_{Z,t})$	Investment condition, robotic capital
$\hat{C}_t^r = \mathbb{E}_t \hat{C}_{t+1}^r - \frac{1}{\gamma} \left[(1 - \beta(1 - \delta_k)) \mathbb{E}_t \hat{r}_{t+1}^K + \beta(1 - \delta_k) \mathbb{E}_t \hat{Q}_{K,t+1} \right]$	Valuation, traditional capital
$\hat{C}_t^r = \mathbb{E}_t \hat{C}_{t+1}^r - \frac{1}{\gamma} \left[(1 - \beta(1 - \delta_z)) \mathbb{E}_t \hat{r}_{t+1}^Z + \beta(1 - \delta_z) \mathbb{E}_t \hat{Q}_{Z,t+1} \right]$	Valuation, robotic capital
$\hat{K}_{t+1} = (1 - \delta_k) \hat{K}_t + \delta_k \hat{I}_{K,t}$	Accumulation, traditional capital
$\hat{Z}_{t+1} = (1 - \delta_z) \hat{Z}_t + \delta_z \hat{I}_{Z,t}$	Accumulation, robotic capital
$\hat{p}_t^Z = \rho_p \hat{p}_{t-1}^Z + \varepsilon_{p,t}$	Relative price process
$\hat{A}_t = \rho_A \hat{A}_{t-1} + \varepsilon_t^A$	Productivity process
$\hat{P}_{L,t} = s_Z \hat{r}_t^Z + (1 - s_Z) \hat{w}_{U,t}$	Unit cost, lower nest
$\hat{Z}_t = \hat{L}_t - \sigma_L (\hat{r}_t^Z - \hat{P}_{L,t})$	Demand for robotic capital
$\hat{N}_{U,t} = \hat{L}_t - \sigma_L (\hat{w}_{U,t} - \hat{P}_{L,t})$	Demand for unskilled labour
$\hat{P}_{X,t} = s_S \hat{w}_{S,t} + (1 - s_S) \hat{P}_{L,t}$	Unit cost, upper nest
$\hat{N}_{S,t} = \hat{X}_t - \sigma_X (\hat{w}_{S,t} - \hat{P}_{X,t})$	Demand for skilled labour
$\hat{L}_t = \hat{X}_t - \sigma_X (\hat{P}_{L,t} - \hat{P}_{X,t})$	Demand for the composite input
$\hat{K}_t + \hat{r}_t^K = \hat{Y}_t + \widehat{mc}_t$	Demand for traditional capital
$\hat{X}_t + \hat{P}_{X,t} = \hat{Y}_t + \widehat{mc}_t$	Demand for the labour-robot bundle
$\widehat{mc}_t = -\hat{A}_t + \alpha \hat{r}_t^K + (1 - \alpha) \hat{P}_{X,t}$	Real marginal cost
$\hat{N}_{S,t} = \nu_{Sr} \hat{N}_{S,t}^r + \nu_{Sh} \hat{N}_{S,t}^h$	Labour aggregation, skilled
$\hat{N}_{U,t} = \nu_{Ur} \hat{N}_{U,t}^r + \nu_{Uh} \hat{N}_{U,t}^h$	Labour aggregation, unskilled
$\hat{C}_t = \chi_r \hat{C}_t^r + \chi_{hs} \hat{C}_{S,t}^h + \chi_{hU} \hat{C}_{U,t}^h$	Aggregate consumption
$\hat{Y}_t = s_C \hat{C}_t + s_{I_K} \hat{I}_{K,t} + s_{I_Z} (\hat{I}_{Z,t} + \hat{p}_t^Z)$	Goods market clearing
$\hat{\pi}_t = \kappa_f \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_p \widehat{mc}_t$	Phillips curve
$\hat{R}_t = \rho_R \hat{R}_{t-1} + (1 - \rho_R) [\phi_\pi \hat{\pi}_t + \phi_x \hat{x}_t] + \varepsilon_t^R$	Monetary policy rule
$\hat{I}_{K,t}^{lag} = \hat{I}_{K,t-1}$	Lag carrier, traditional investment
$\hat{I}_{Z,t}^{lag} = \hat{I}_{Z,t-1}$	Lag carrier, robotic investment
$\hat{R}_t^{lag} = \hat{R}_{t-1}$	Lag carrier, policy rate

Table 6: The log-linear system. Seven predetermined variables, twenty-six jump variables.

The Phillips curve is coded with $\zeta_p = 0$ imposed, so $\kappa_f = \beta$ and no lagged inflation term or additional state appears. The policy rule responds to the output gap $\hat{x}_t = \hat{Y}_t - \hat{Y}_t^*$; since a monetary innovation leaves the flexible-price allocation unchanged, $\hat{Y}_t^* = 0$ and the gap coincides with output in every experiment reported, so the rule is implemented in $\hat{Y}_t$. The distinction would matter for a productivity or relative price shock, and the flexible-price block required for that case is not part of the system solved here.

B.4 Solution method

The system is written as $A\mathbb{E}_t x_{t+1} = Bx_t + C\varepsilon_t$, with x_t ordered predetermined variables first, and solved by the method of Klein (2000). A generalised Schur decomposition of the pencil (B, A) is computed with the stable eigenvalues ordered first, which yields the transition matrix for the predetermined block and the policy function mapping states into jump variables. The Blanchard–Kahn condition is checked as an assertion at every solve: the number of eigenvalues inside the unit circle must equal the number of predetermined variables, seven, and a violation raises an error rather than returning a solution. Because the decomposition is taken of (B, A) in that order, the generalised eigenvalues reported by the routine are already the ratios required by the method and no further inversion is needed.

Impulse responses are computed by solving jointly for the impact effect of each innovation on the jump variables and on next period’s predetermined block, then iterating the transition forward with no further innovations. Responses are reported as percentage deviations from the steady state of the configuration in question, so that each regime is compared against its own steady state rather than a common one.

Because the same structural innovation delivers a different movement in the policy rate across automation regimes, impulse responses are normalised per regime to a common impact response of the policy rate of twenty-five basis points, quarterly, rather than to a common innovation. This compares regimes at a common policy action. The alternative of a single scaling factor computed from one reference regime is not neutral: under that convention the aggregate consumption ranking across regimes reverses, so the choice of normalisation is reported here rather than left implicit.

The model is solved in Python. The replication package at https://github.com/alvaromezquita/robot_tank_model contains the steady-state solver, the linear system and the Klein solution routine, together with scripts reproducing every table and figure in the paper and in the appendices.

C Robustness

C.1 Frictionless investment

This subsection of the appendix reports the same experiment with investment adjustment costs removed, setting $\tau_K = \tau_Z = 0$ and leaving every other parameter at its baseline value. The steady state is unaffected, since the adjustment cost function and its first derivative vanish at the deterministic steady state by construction; the amplification factor of Table 3 is therefore identical in the two specifications, which provides a useful check on the numerical solution.

The aggregate result is robust. The sacrifice ratio falls from 3.20 at $\Omega_Z = 0.05$ to 1.20 at $\Omega_Z = 0.35$, a decline of sixty-three per cent, against fifty-four per cent in the baseline. The steepening of the inflation–output trade-off with automation is therefore not an artefact of the investment friction, which is what one would expect given that the mechanism operates through the predeterminacy of the robot stock rather than through the cost of changing it. The distributional ordering is likewise unchanged: constrained households bear a larger consumption loss than savers at every level of automation, unskilled constrained households bear more than skilled, and both gaps widen in the robot income share.

The levels are a different matter, and the comparison is between specifications rather than against the data. Normalised to the same twenty-five basis point impact response of the policy rate, output contracts by nearly forty per cent on impact in the least automated regime under the frictionless specification, against 1.4 per cent in the baseline: a given monetary tightening is roughly twenty-nine times more powerful once the cost of adjusting the stocks is removed. The absolute magnitude of transmission in the baseline is itself high relative to the empirical literature, for the reasons set out in Section 4.1, so the point here is not that one specification is empirically plausible and the other is not. It is that without an adjustment cost the capital and robot margins absorb the entire disturbance instantaneously, and the resulting transmission is violent enough that the specification cannot be used to say anything about magnitudes at all.

The dynamics make the source of the problem explicit. Both stocks fall below their steady-state values from the quarter after impact, since the collapse in investment is large enough to move them materially in a single period. Lower stocks raise real marginal cost, so inflation turns positive from the second quarter onwards while output remains below trend, and the interest rate rule responds by holding the policy rate elevated: it decays at 0.96 per quarter in this specification, against 0.37 in the baseline. A contractionary demand disturbance is thereby converted into a persistent supply contraction, with the sign of the inflation response reversing within two quarters. Nothing in this is inconsistent, but it is not a description of monetary transmission that any empirical account would recognise, and it is the reason the baseline specification carries adjustment costs at the curvature estimated by [Smets and Wouters \(2003\)](#).

C.2 Sensitivity to the elasticity of substitution between robots and unskilled labour

The elasticity of substitution in the lower nest, σ_L , governs how readily robotic capital replaces unskilled labour, and the baseline sets it at 10. No empirical study provides a value that can be transported directly into this specification, so this appendix reports the experiment of Section 4 at $\sigma_L \in \{0.95, 2, 10\}$, spanning near-complementarity, moderate substitutability and the baseline. All other parameters are held at their baseline values and the calibration targets are unchanged, so θ_Z and θ_S are re-solved at each value.

The difficulty is one of nesting rather than of data. [Battisti et al. \(2021\)](#) estimate the relevant elas-

ties on industry panels, but within a structure that pairs robotic capital with skilled labour and the resulting composite with unskilled labour, so neither of their estimates corresponds to the substitution margin between robots and unskilled labour that σ_L governs here. Their estimates do bracket the baseline in order of magnitude, and support the ordering the model imposes, but they cannot pin the parameter down.

Table 7 reports the results. The aggregate result is insensitive. The sacrifice ratio falls from 1.65 to 0.73 at $\sigma_L = 0.95$ and from 1.68 to 0.76 at $\sigma_L = 10$, the two differing by no more than four per cent at any regime. This is less surprising than it first appears. The amplification factor of Table 3 is identical at every value of σ_L , because the cost share of robots in the lower nest is pinned by the robot income share target rather than by the elasticity; and it is that cost share, together with the predeterminacy of the robot stock, which drives the steepening of the inflation–output trade-off. The mechanism of Section 4.4 is therefore invariant to this parameter by construction rather than by numerical accident.

The excess burden borne by constrained households is also largely insensitive. For unskilled constrained households it widens from -0.046 to -0.088 across the regimes at $\sigma_L = 0.95$, against -0.052 to -0.117 at the baseline, so the lowest value considered retains between three-quarters and seven-eighths of the baseline magnitude. The asset-market participation channel does not run through the lower nest, and the elasticity accordingly has little to say about it.

What the elasticity does govern is the skill dimension. The gap between unskilled and skilled constrained households falls from between fifteen and thirty-eight per cent of the skilled loss at the baseline to between two and three and a half per cent at $\sigma_L = 0.95$. This is the expected result: when robots and unskilled labour are near-complements, the composition of the lower nest is close to fixed and unskilled labour is no longer the margin on which adjustment falls. The magnitude of the skill result reported in Section 4.5 should therefore be read as conditional on substitutability, while its sign and its monotonicity in the robot income share are not.

The relationship is strongly concave. At $\Omega_Z = 0.05$ the gap rises from 2.3 per cent of the skilled loss at $\sigma_L = 0.95$ to 9.1 per cent at $\sigma_L = 2$, and only to 14.6 per cent at $\sigma_L = 10$. Almost all of the variation occurs below a value of two, and the result is close to flat over the upper part of the range. The choice of 10 is accordingly not a choice of magnitude so much as a choice of region: any value in the upper range delivers substantially the same answer.

A final observation bears on how the low end of the range should be interpreted. The steady state degenerates as σ_L falls and automation rises. At $\sigma_L = 0.95$, attaining a robot income share of 0.35 requires a robot stock almost four hundred times larger than at the baseline, with steady-state output of 595 against 1.5 and a robot share parameter of 0.954. When robots and unskilled labour are complements, the marginal product of robots falls steeply in the stock, and a high robot income share can only be reached by accumulating an implausible quantity of them. The low- σ_L economies at high automation intensity are therefore solutions to the calibration problem rather than descriptions of an economy, and they are reported here for completeness rather than as alternative baselines. This is a further reason for locating the benchmark in the substitutable

region.

Table 7: Sensitivity of the principal results to the elasticity of substitution between robots and unskilled labour.

Robot income share Ω_Z	0.05	0.15	0.25	0.35
<i>Panel A: sacrifice ratio, $\hat{y}/\hat{\pi}$</i>				
$\sigma_L = 0.95$	1.650	1.327	1.024	0.734
$\sigma_L = 2$	1.665	1.349	1.043	0.749
$\sigma_L = 10$	1.678	1.369	1.062	0.764
<i>Panel B: excess burden, unskilled constrained households, $\hat{C}_U^h - \hat{C}^r$</i>				
$\sigma_L = 0.95$	-0.0463	-0.0534	-0.0649	-0.0879
$\sigma_L = 2$	-0.0498	-0.0605	-0.0752	-0.1029
$\sigma_L = 10$	-0.0525	-0.0670	-0.0848	-0.1172
<i>Panel C: skill gap among constrained households, $\hat{C}_U^h - \hat{C}_S^h$</i>				
$\sigma_L = 0.95$	-0.0013	-0.0018	-0.0023	-0.0033
$\sigma_L = 2$	-0.0051	-0.0097	-0.0137	-0.0198
$\sigma_L = 10$	-0.0082	-0.0169	-0.0244	-0.0355

C.3 Sensitivity to relative risk aversion

The coefficient of relative risk aversion is set below unity in the benchmark, at $\gamma = 0.75$. The value matters because unity is a boundary in this model rather than merely a point on a continuum: the hours response of constrained households carries the elasticity $(1 - \gamma)/(\varphi + \gamma)$, which changes sign there. This appendix reports the experiment of Section 4 at $\gamma = 1.25$, with all other parameters at their benchmark values and the calibration targets unchanged.

The qualitative change is confined to the labour supply of constrained households, and it is the change the boundary predicts. At $\gamma = 0.75$ the substitution effect dominates and constrained hours fall with the wage; at $\gamma = 1.25$ the income effect dominates and constrained hours rise, since a household consuming its current labour income in full must work more when that income falls. At unity the two offset exactly and constrained hours do not respond at all. Table 8 reports the reversal.

Nothing else changes qualitatively. The consumption of constrained households falls with the wage at either value, because the consumption elasticity $(1 + \varphi)/(\varphi + \gamma)$ is positive throughout; the reversal in hours changes its magnitude but not its sign. The aggregate result is likewise unaffected in the dimension the paper claims: the sacrifice ratio declines by fifty-two per cent across the regimes at $\gamma = 1.25$, against fifty-four per cent at the benchmark.

What γ does govern is the scale of every response in the model. Raising it from 0.75 to 1.25 reduces

the impact response of Ricardian consumption by a factor of 0.61, against an intertemporal substitution factor of 0.60, and the remaining responses are scaled in close proportion. This is a statement about the magnitude of monetary transmission rather than about its distributional structure, and it leaves the relative magnitudes reported in Section 4.5 nearly intact: the gap between unskilled and skilled constrained households runs from fourteen to thirty-three per cent of the skilled loss at $\gamma = 1.25$, against fifteen to thirty-eight per cent at the benchmark.

The parameter is nevertheless worth flagging rather than burying, because the benchmark value is not identified by the aggregate moments used to discipline the model. The data employed here pin down the scale of the consumption response but not the sign of the constrained employment response, since the two are governed by different combinations of the same preference parameters. The consumption results of Section 4.5 are robust to this; the employment margin is not, and claims about hours should be read as conditional on the benchmark value. Leduc and Liu (2024) provide the relevant case: with bargained wages and a procyclical automation threat, the unskilled wage responds through the value of the firm’s outside option rather than through the curvature of household utility, and employment adjusts on a matching margin that the present specification does not contain.

Table 8: Sensitivity of the principal results to relative risk aversion.

Robot income share Ω_Z	0.05	0.15	0.25	0.35
Panel A: sacrifice ratio, $\hat{y}/\hat{\pi}$				
$\gamma = 0.75$	1.678	1.369	1.062	0.764
$\gamma = 1.25$	1.302	1.074	0.846	0.621
Panel B: excess burden, unskilled constrained households, $\hat{C}_U^h - \hat{C}^r$				
$\gamma = 0.75$	−0.0525	−0.0670	−0.0848	−0.1172
$\gamma = 1.25$	−0.0348	−0.0440	−0.0564	−0.0811
Panel C: skill gap among constrained households, as a share of the skilled loss				
$\gamma = 0.75$	14.6%	27.4%	33.8%	38.0%
$\gamma = 1.25$	13.8%	24.0%	29.4%	33.0%
Panel D: hours of unskilled constrained households, $\hat{N}_U^h$				
$\gamma = 0.75$	−0.0062	−0.0075	−0.0092	−0.0123
$\gamma = 1.25$	0.0040	0.0049	0.0061	0.0084

C.4 Sensitivity to the share of constrained households

C.4.1 The participation margin

The benchmark sets the share of households excluded from asset markets at $\lambda = 0.24$. This appendix reports the experiment of Section 4 at $\lambda \in \{0.10, 0.30\}$, which spans the range of estimates

in the literature, with all other parameters at their benchmark values and the calibration targets unchanged. Table 9 reports the results.

Two properties hold across the range and they are the properties on which Section 5.2 relies. The first is that the gradient of the aggregate result in the robot income share is invariant to the participation share. The sacrifice ratio declines by 54.2 per cent across the full grid at $\lambda = 0.10$, by 54.4 per cent at the benchmark and by 54.2 per cent at $\lambda = 0.30$. The level of the sacrifice ratio rises with λ , which is the familiar amplification of transmission by limited participation, but the rate at which automation lowers it does not.

The second is that the relative distributional structure is likewise invariant. The gap between unskilled and skilled constrained households, expressed as a share of the skilled loss, runs from 14.0 to 39.4 per cent at $\lambda = 0.10$ and from 15.0 to 37.1 per cent at $\lambda = 0.30$. The composition of the burden among constrained households is therefore a function of automation intensity and not of how many households are constrained.

What does depend on the participation share is the level of the distributional cost, and it does so more than proportionally. At $\Omega_Z = 0.05$ the excess burden borne by unskilled constrained households is -0.031 , -0.053 and -0.078 at $\lambda = 0.10$, 0.24 and 0.30 respectively: the increment from the benchmark to 0.30 exceeds the increment from 0.10 to the benchmark, over less than half the distance. The burden is convex in the participation share, as the aggregation of consumption responses with λ weights implies.

The two margins also interact, although mildly. The factor by which the excess burden rises between successive values of λ is itself increasing in the robot income share, from 1.72 to 1.90 between $\lambda = 0.10$ and the benchmark, and from 1.48 to 1.72 between the benchmark and $\lambda = 0.30$. Automation intensity and limited participation are therefore complements in the level of the distributional cost, not independent contributors to it. They remain separable in the sense that matters for Section 5.2: one governs the level and the other the gradient.

C.4.2 Full asset market participation

Setting $\lambda = 0$ removes the constrained households entirely and returns the model to a representative-agent economy, retaining the nested production structure and the predetermined robot stock. The comparison isolates what the aggregate results owe to limited participation, which is worth establishing directly, since the amplification of monetary transmission by hand-to-mouth households is a well-documented channel and a reader is entitled to suspect that the steepening documented in Section 4.3 is a version of it.

It is not. With no constrained households in the economy, the sacrifice ratio falls from 1.486 at $\Omega_Z = 0.05$ to 0.686 at $\Omega_Z = 0.35$, a decline of 53.9 per cent across the full grid, against 54.4 per cent in the benchmark specification over the same range.

The role of limited participation is accordingly one of amplification rather than of generation. The wage divergence is produced by the nesting and the predetermined stock; the constraint converts it into a consumption divergence and roughly doubles it. This division is what permits the two results of Section 4 to be read as one, as Section 5.1 argues, while still allowing the participation margin to be addressed independently of the automation margin.

Table 9: Sensitivity of the principal results to the share of households excluded from asset markets.

Robot income share Ω_Z	0.05	0.15	0.25	0.35
Panel A: sacrifice ratio, $\hat{y}/\hat{\pi}$				
$\lambda = 0$	1.486	1.217	0.948	0.686
$\lambda = 0.10$	1.538	1.257	0.976	0.704
$\lambda = 0.24$	1.678	1.369	1.062	0.764
$\lambda = 0.30$	1.799	1.470	1.142	0.824
Panel B: relative wage response, $\hat{w}_U - \hat{w}_S$				
$\lambda = 0$	-0.0038	-0.0080	-0.0112	-0.0153
$\lambda = 0.10$	-0.0047	-0.0097	-0.0137	-0.0188
$\lambda = 0.24$	-0.0075	-0.0152	-0.0220	-0.0321
$\lambda = 0.30$	-0.0105	-0.0219	-0.0329	-0.0522
Panel C: skill gap among constrained households, as a share of the skilled loss				
$\lambda = 0$	13.4%	26.8%	34.2%	40.1%
$\lambda = 0.10$	14.0%	27.1%	34.2%	39.4%
$\lambda = 0.24$	14.6%	27.4%	33.8%	38.0%
$\lambda = 0.30$	15.0%	27.3%	33.4%	37.1%

C.5 Without interest rate smoothing

Setting the interest rate smoothing parameter to zero removes the endogenous persistence of the policy rate and leaves the Taylor rule responding only to contemporaneous inflation and output. The shock is renormalised in each regime to deliver the same twenty-five basis point impact response of the policy rate, as in the baseline, and the policy rate returns to steady state by the first quarter after the shock. Table 10 reports the resulting trade-off.

The gradient survives and is steeper than in the baseline. The impact output cost of disinflation falls by forty-one percent over the range of robot income shares from five to twenty-five percent, against thirty-seven percent with the baseline degree of smoothing. Smoothing therefore damps the gradient rather than generating it, and the specification adopted in the main text is the conservative one. The mechanism is unchanged, since (29) concerns the factor mix and not the conduct of policy.

The exercise also bears on the object being measured. Without smoothing the response is heavily

front-loaded: ninety percent of the cumulative output loss falls in the impact quarter at the lowest robot share. Reporting the cumulative output loss per unit of cumulative disinflation rather than the ratio of impact responses therefore changes the level of the statistic but not the comparison, and the cumulative measure declines by thirty-six percent over the same range. The share of the output loss borne in the impact quarter itself falls from ninety to seventy-eight percent across the grid, so automation makes the output response modestly more drawn out even as it makes the total smaller.

Table 10: Trade-off without interest rate smoothing

Ω_Z	0.05	0.15	0.25	0.35
Impact output cost, $\hat{y}/\hat{\pi}$	2.70	2.14	1.59	1.08
Cumulative output cost	3.05	2.53	1.96	1.37
Share of output loss at impact	0.90	0.86	0.82	0.78